CIRCUIT STUDY GUIDE · 0.18 µm CMOS · VIVA PREPARATION

Adaptive-Bias Capless LDO

Every transistor, every wire, and every number in your schematic — built up from what a MOSFET is, so you can defend any connection an examiner points at. Nothing here is assumed known except voltage, current, and "a transistor is a switch-like device".

V_{IN} **1.8 V**	V_{OUT} **1.6 V**	DROPOUT **200 mV**
I_{LOAD} **0–100 mA**	C_L **100 pF**	DEVICES **28 MOS**

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PART 00

Read this first — what your brief asks for, and what actually applies

Two honest corrections before we start, so you don't walk into a viva carrying a wrong frame.

CORRECTION 1 — THERE IS NO PLC ANYWHERE IN THIS CIRCUIT

Your question list asks for PLC basics, sourcing vs sinking, relay vs transistor outputs, scan cycles, and where the PLC fits into your circuit. **Nothing in this schematic is a PLC, and a PLC cannot be connected to it.** That part of the brief belongs to a different kind of project (industrial control panel wiring) and was almost certainly carried over from a generic template.

So you know the term anyway: a **PLC (Programmable Logic Controller)** is a ruggedised industrial computer — a box on a DIN rail in a factory panel — that reads sensor inputs, runs a small program in a repeating "scan cycle", and switches outputs like motors and valves. It works at 24 V, at speeds of milliseconds, with wires you can see.

Your circuit is an **analog integrated circuit**: it lives *inside* a silicon chip, works at 1.8 V, responds in nanoseconds, and has no wires you can touch. The smallest device in it is roughly one ten-thousandth the width of a human hair. A PLC is to this circuit what a shipping container is to a grain of rice — not a bigger version of the same idea, a different category of object entirely.

If an examiner asks "what controls this circuit?" the correct answer is: *nothing external controls it moment-to-moment.* It is a self-regulating analog feedback loop. The only things fed in from outside the block are a reference voltage V_{REF} (from a bandgap reference circuit elsewhere on the same chip) and a bias current I_{BIAS} (from a current reference on the same chip). A digital controller on the chip might switch the whole LDO on or off through an enable pin, but that pin isn't drawn here.

CORRECTION 2 — BOARD-LEVEL ITEMS DON'T EXIST ON-CHIP

Your brief also asks for pull-up/pull-down resistor values, gate resistors, heatsinking, and fuses. On a printed circuit board those are all real. **Inside this chip there is not one resistor in the schematic** — that is deliberate, and it is a point worth making in your viva (resistors are enormous and inaccurate in CMOS; every biasing job that a resistor would do on a board is done here by a transistor instead). There is no gate resistor because there is no wire inductance to damp — the "wire" from the amplifier to the pass transistor's gate is a few hundred micrometres of metal. There is no fuse; the on-chip equivalent is a *current-limit circuit*, which this schematic does not have, and Part 17 shows you how to add one as your own contribution.

How to use this guide

Parts 01–04 build the physics from zero. Part 05 is the single most valuable page for a viva: the complete netlist, every terminal of every transistor. Parts 06–10 walk each coloured block of your schematic. Parts 11–13 are tracing and numbers. Parts 14–17 are judgement, failure modes and questions.

Throughout, node names are written like V_G and device names like M_{P6}, matching your schematic exactly. The three coloured blocks keep their colours: **error amplifier (red)**, **adaptive bias (blue)**, **slew-rate enhancement (green)**.

PART 01

What an LDO is, and why yours looks complicated

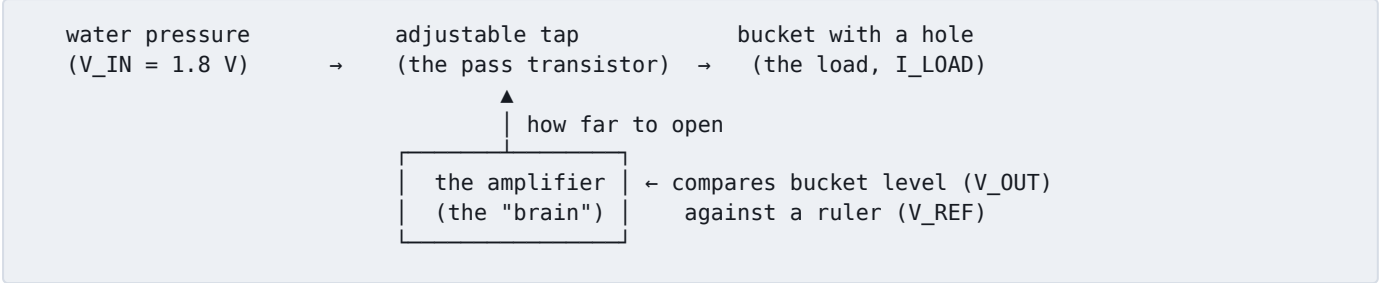
Start with the job. Everything in the schematic exists to do this one job better.

The job

A chip has one battery or one main supply — call it 1.8 V. But 1.8 V from a battery is not really 1.8 V. It sags when the phone's radio transmits, it has noise on it, it drifts as the battery discharges. Meanwhile a sensitive block on the chip — an ADC, an oscillator, a memory — needs a *clean, steady* 1.6 V that does not move when anything else on the chip does.

A **voltage regulator** is the circuit that takes the messy supply and produces the clean one. A **Low-DropOut regulator (LDO)** is a regulator that can do this even when the input is only *slightly* above the output. The gap it needs is called the **dropout voltage**. Yours is 200 mV: give it 1.8 V in, it gives 1.6 V out.

The simplest possible mental model: a tap



You want the water level in the bucket held at exactly one mark. Someone watches the level, compares it to the mark, and adjusts the tap continuously. If the hole gets bigger (load draws more current), the level starts to fall, the watcher opens the tap further, and the level comes back. That is **negative feedback**, and it is the whole principle of your circuit.

Three pieces map directly onto your schematic:

- **The tap** = M_P, the big pass transistor in the middle of the page.
- **The ruler (the mark to hold)** = V_{REF}, generated elsewhere on the chip.
- **The watcher** = the **error amplifier** in the red box, which measures the difference between V_{OUT} and V_{REF} and produces the tap-opening command V_G.

An LDO throws away the excess as heat — it does not convert it

This is the honest limitation, and examiners love it. The pass transistor is a controlled resistor in series with the load. All the load current flows through it, and it drops ($V_{IN} - V_{OUT}$) volts. That product is power turned into heat:

$$P_{loss} = (V_{IN} - V_{OUT}) \times I_{LOAD} = (1.8 - 1.6) \times 0.1\ A = \mathbf{20\ mW}$$

$$\text{Efficiency}\quad \eta = V_{OUT} \cdot I_{LOAD} \,/\, (V_{IN} \cdot (I_{LOAD} + I_Q))$$
$$= (1.6 \times 0.1) \,/\, (1.8 \times 0.10003) = \mathbf{88.9\ \%} \quad \leftarrow \text{ceiling is } V_{OUT}/V_{IN}$$

An LDO's efficiency can never exceed V_{OUT}/V_{IN}. Drop 1.6 V from 3.3 V and you burn more than half the energy. That is why LDOs are used for *small, quiet* loads, and switching regulators (buck converters) are used for big ones — but a buck converter makes electrical noise from its switching, which is exactly what the sensitive block was trying to escape. So real chips use a buck to get roughly close, then an LDO to clean it up.

Why yours has 28 transistors instead of about 8

A textbook LDO needs a pass transistor, a five-transistor amplifier, and a feedback divider. Yours has three extra sub-circuits. Each one exists to defeat a specific problem, and being able to name the problem is the difference between a pass and a distinction:

BLOCK IN YOUR SCHEMATIC	THE PROBLEM IT DEFEATS
Cross-coupled pair M_{P3}, M_{P4}	A single-stage amplifier running on 1 µA doesn't have enough gain to regulate tightly. These add positive feedback that boosts gain without adding a second stage (which would add a pole and hurt stability).
Doubled input pairs M_{N1}–M_{N4}	Gives a fast, direct path from the inputs to the output node that skips the current mirror's delay, and lets the output stage both push and pull hard.
Adaptive bias M_{P9}, M_{N10}, M_{N9}	You want low quiescent current at light load (battery life) but high amplifier speed at heavy load. Those conflict. This block senses the load and raises the amplifier's bias current only when the load needs it.
Slew-rate enhancement M_{P10}–M_{P15}, M_{N11}, M_{N12}	The pass transistor's gate is a ~7 pF capacitor. At 2 µA of amplifier current it takes **8 microseconds** to swing that gate. This block dumps hundreds of microamps into the gate on a fast edge, cutting it to ~50 ns — while burning almost no current at rest.
No output capacitor	This is an **output-capacitorless (capless) LDO**: C_L is only 100 pF of on-chip parasitic, not a 1 µF part on a board. That saves a pin and a component, but removes the energy reservoir that normally absorbs load steps — which is precisely why the two blocks above are needed.

> THE ONE-SENTENCE ANSWER
>
> "It is an output-capacitorless LDO in which a cross-coupled current-mirror error amplifier drives a PMOS pass device, with load-dependent adaptive biasing to trade quiescent current against bandwidth, and a capacitively-coupled slew-rate booster to overcome the pass transistor's large gate capacitance during load transients."

PART 02

DC, polarity, ground and rails

Short section, because on-chip the answers are simpler than on a board — but they are still asked.

DC vs AC

DC (direct current) means the voltage holds one polarity and, ideally, one value: the top wire is always positive with respect to the bottom wire. **AC (alternating current)** means the polarity reverses periodically — mains at 50 Hz swings +325 V to −325 V fifty times a second.

Your circuit is DC because **MOSFETs are inherently one-polarity devices**. A PMOS transistor conducts when its source is the more positive terminal; an NMOS when its source is the more negative one. Every device in the schematic is drawn assuming `VIN` is above `GND`. Apply AC and on the negative half-cycle every transistor's source and drain swap roles, every bias current collapses, and the parasitic diodes formed by the wells (see Part 04) forward-bias and short the chip internally. The chip is destroyed, not merely inoperative.

Rails, ground and polarity in this schematic

```
VIN  ──────────────────────────────────────  1.8 V   ← the positive supply rail
                                                     (top wire, runs all the way across)
       everything happens between these two

GND  ──────────────────────────────────────  0.0 V   ← the reference / return rail
                                                     (bottom wire, "common ground")
```

- **Ground** is not a magic sink. It is simply the node we *declare* to be 0 V, because voltage is only ever a *difference* between two points. Every other voltage on the page — V_{OUT} = 1.6 V, V_G, V_{AB} — means "with respect to GND".
- **Common ground** means the amplifier, the load, the reference and the bias circuit all measure against the *same* 0 V. If the load returned to a different ground that sat 50 mV higher, the amplifier would regulate V_{OUT} to 1.6 V relative to *its* ground while the load saw 1.55 V. On a chip this shows up as **ground bounce** from IR drop in the metal — which is why the 100 mA return path must be a wide, low-resistance metal bus, and why the sensitive amplifier ground should connect to the pad separately from the pass transistor's ground. That is a real layout answer, not a theoretical one.
- **Conventional current** flows from VIN, through the transistors, into the load, and back to GND. Electrons physically go the other way; ignore this, all circuit analysis uses conventional current.

What happens if V_{IN} is wrong

CONDITION	CONSEQUENCE	WHY
V_{IN} < 1.8 V, still > ~1.8 V	Normal regulation	Loop has headroom; V_G simply sits lower.
V_{IN} → 1.7 V at 100 mA	**Dropout**: V_{OUT} falls to ≈1.5 V and tracks V_{IN}	Loop saturates — V_G is already at its lowest, M_P is fully on and behaving as a plain resistor. Regulation is lost; output follows input minus $I{\cdot}R_{DS(on)}$.
V_{IN} < ~1.1 V	Circuit stops working entirely	Not enough headroom to stack a V_{GS} (0.5 V) plus a V_{DSsat} (0.15 V) plus the tail device in the amplifier. Branches fall out of saturation.

CONDITION	CONSEQUENCE	WHY
V_{IN} > ~2.0 V	Works, but worse; above ~2.5 V, permanent damage	Efficiency drops and the pass device heats more. Beyond the process limit the 4.1 nm gate oxide breaks down — a one-way failure. 0.18 µm thin-oxide devices are rated for 1.8 V ±10 %.
Reverse polarity	Immediate destruction	Every n-well-to-substrate junction forward-biases; the chip becomes a short across your supply.

The last two rows are exactly why real products put an **under-voltage lockout (UVLO)** and an **over-voltage clamp** around an LDO. Neither is in your schematic. Note that as a limitation — it is a genuine one.

PART 03

The MOSFET, from zero

Twenty-eight of the twenty-nine components in your schematic are MOSFETs. Everything below is asked in vivas almost verbatim.

What a transistor is

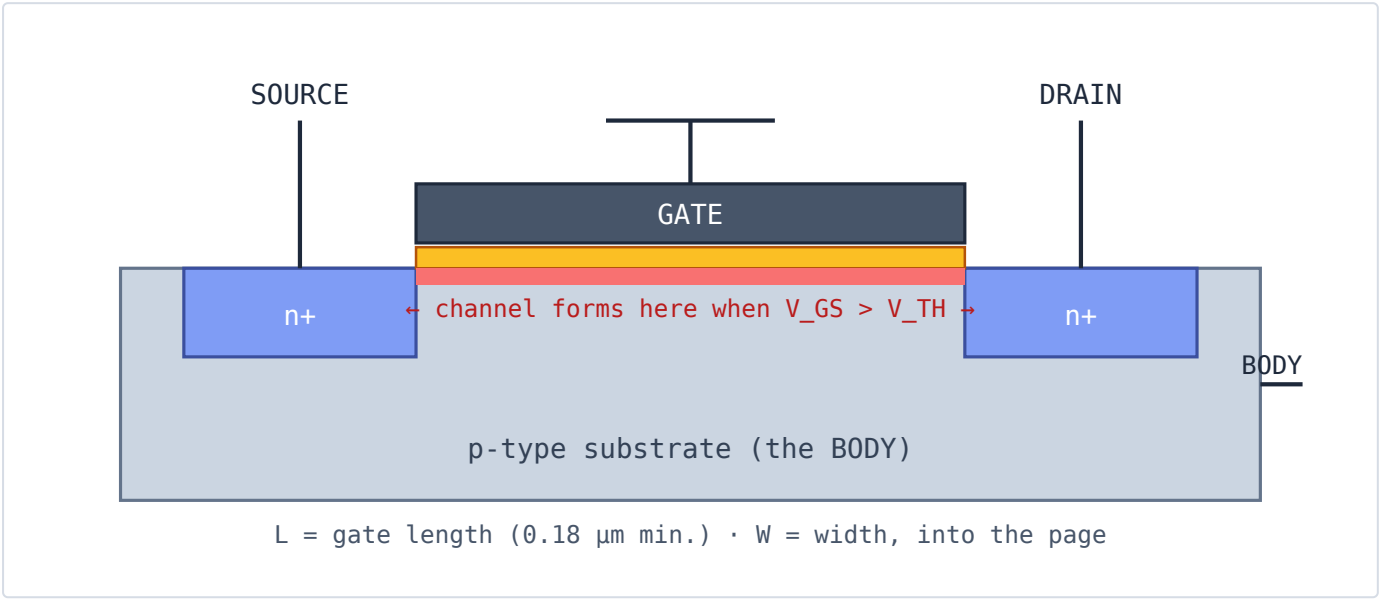
A **transistor** is a three-terminal device in which the voltage or current at one terminal controls the current flowing between the other two. That's it. It is an electrically-operated valve. Because a small input can control a large output, transistors can either **switch** (fully open or fully shut) or **amplify** (partly open, in proportion to the input).

What a MOSFET is

MOSFET = **M**etal-**O**xide-**S**emiconductor **F**ield-**E**ffect **T**ransistor. Read the name backwards and it explains itself:

- **Field-Effect**: it is controlled by an electric *field* (i.e. a voltage), not by an input current.
- **Metal-Oxide-Semiconductor**: the physical sandwich. A conductor (the gate, actually polysilicon in modern CMOS) sits on top of an **insulator** (silicon dioxide, glass — only 4.1 nm thick in your 0.18 µm process, about 20 atoms), which sits on the silicon.

The key structural fact: **the gate is separated from everything else by an insulator.** No current can flow into the gate in steady state. The gate is one plate of a very small capacitor.



N-channel MOSFET cross-section. Two n-type islands sit in p-type silicon. Left alone they cannot pass current — but a positive gate voltage pulls electrons up to the surface and builds a conducting bridge between them.

Gate, Drain, Source — what each name means

TERMINAL	SYMBOL	WHAT IT PHYSICALLY IS	ITS JOB
Gate	G	Conducting plate over the insulator	The control knob. Its voltage decides how much the valve opens. Draws no DC current.
Source	S	One of the two doped islands	Where the charge carriers come *from*. All voltages are measured relative to it — the gate's control is V_{GS}, gate *minus source*.
Drain	D	The other doped island	Where the carriers *go*. Current flows S→D inside an NMOS channel (electrons D→S).
Body	B	The silicon underneath (substrate or well)	The fourth terminal, usually tied to the most negative point (NMOS→GND) or most positive (PMOS→VIN). Rarely drawn, always present.

> WHICH ISLAND IS THE SOURCE?
>
> Source and drain are physically identical. The *circuit* decides which is which: **for an NMOS the source is whichever terminal is at the lower voltage; for a PMOS it is whichever is at the higher voltage.** This is a favourite viva question and it matters in Part 04, where a MOSFET's roles can swap and it starts behaving like a diode.

N-channel vs P-channel

NMOS (N-channel)

- Carriers are **electrons** — fast ($\mu_n \approx 417$ cm^2/V·s in your process).
- Source goes to the **lower** voltage, usually GND.
- Turns **ON** when V_{GS} is **positive** and above V_{TH} (≈ +0.50 V here).
- Good at **pulling a node down** to ground.
- In your schematic: all the $M_{N...}$ devices — the input pairs, tail sources, and mirror M_{N7}/M_{N8}.

PMOS (P-channel)

- Carriers are **holes** — about 5× slower ($\mu_p \approx 85$ cm^2/V·s), so a PMOS needs ~5× the width for the same current.
- Source goes to the **higher** voltage, usually VIN.
- Turns **ON** when V_{GS} is **negative** — easier to say as $V_{SG} > |V_{TH}|$ (≈ 0.51 V here).
- Good at **pulling a node up** to the supply.
- In your schematic: all the $M_{P...}$ devices, including the big pass transistor M_P.

The little arrow on the symbol tells you which is which. In your schematic every PMOS has its bubble/arrow at the top touching the VIN rail; every NMOS sits with its source on GND or on a tail node.

Enhancement vs depletion

Every device in your circuit is an **enhancement**-mode MOSFET: it is OFF at $V_{GS} = 0$ and you must "enhance" it with gate voltage to turn it on. **Depletion**-mode devices are ON at $V_{GS} = 0$ and must be pushed off — they are not available in a standard CMOS process and do not appear here. If asked, that one sentence is the complete correct answer.

How a MOSFET actually turns on — threshold voltage

Take the NMOS in the picture. At $V_{GS} = 0$, the two n-islands sit in p-type silicon with no path between them: no current, whatever you do at the drain.

Raise the gate voltage. The gate is one plate of a capacitor; positive charge on it repels the p-type material's holes away from the surface and attracts the few free electrons towards it. At first this only clears a depletion region. Then, at a particular gate voltage, so many electrons have gathered at the surface that the thin layer there stops behaving like p-type silicon and starts behaving like n-type: an **inversion layer**. Now there is a continuous n-type bridge from source to drain, and current can flow.

The threshold voltage V_{TH} is the gate–source voltage at which that inversion layer forms. In your process $V_{TH,n} \approx 0.50$ V and $|V_{TH,p}| \approx 0.51$ V.

> THE COMMON STUDENT MISTAKE
>
> "Below V_{TH} the transistor is off" is only approximately true. Below threshold the current doesn't stop — it falls *exponentially*, roughly ×10 for every 80 mV. This is **subthreshold** or **weak inversion** operation, and it is not a defect: **your error amplifier deliberately runs there**, because at 0.5 µA per device it is the only way to get useful transconductance from 1 µA of bias. Being able to say this is a distinction-level answer.

V_{GS} decides how much; V_{DS} decides which region

Two different voltages do two different jobs, and confusing them is the single most common error.

```
V_GS  ─── sets HOW MUCH current the device wants to pass  (the valve setting)
V_DS  ─── sets WHICH REGION it operates in                (how it behaves)
```

Region	Condition (NMOS)	Behaviour	Used in your circuit for
Cut-off	$V_{GS} < V_{TH}$	essentially no current	devices idle / SRE at rest
Triode (linear)	$V_{GS} > V_{TH}$ and $V_{DS} < V_{GS} - V_{TH}$	acts like a RESISTOR $R = R_{DS(on)}$	pass device M_P in dropout, switches
Saturation (active)	$V_{GS} > V_{TH}$ and $V_{DS} > V_{GS} - V_{TH}$	acts like a CURRENT SOURCE controlled by V_{GS}	every amplifier and mirror device in the red box

The quantity $V_{GS} - V_{TH}$ appears so often it has a name: the **overdrive voltage** V_{OV}, sometimes V_{DSsat}. It is "how far past the threshold you have pushed the gate".

Saturation (the region every amplifier device works in):

$$I_D = \tfrac{1}{2} \cdot \mu_n C_{ox} \cdot (W/L) \cdot (V_{GS} - V_{TH})^2 \cdot (1 + \lambda V_{DS})$$

Triode (the region the pass device works in at full load):

$$I_D = \mu_p C_{ox} \cdot (W/L) \cdot \left[(V_{SG} - |V_{TH}|) \cdot V_{SD} - V_{SD}^2/2 \right]$$

Subthreshold (where your error amplifier lives):

$$I_D \propto \exp\left(V_{GS} / (n \cdot V_T) \right) \quad \text{with } n \approx 1.3, \; V_T = kT/q = 25.9 \text{ mV at } 27\,^\circ\text{C}$$

W/L is the only thing a designer chooses. μC_{ox}, V_{TH} and λ are fixed by the fabrication process. W is the transistor's width, L its length; the ratio sets how much current the device passes for a given gate voltage. Designing this circuit *is* choosing 28 values of W/L.

Gate current is (almost) zero — and why that matters here

The gate sits on an insulator, so in steady state $\mathbf{I_G = 0}$. Two consequences, both directly relevant to your circuit:

1. **Good news:** the error amplifier needs no DC current to hold the pass transistor open. That is why the LDO's quiescent current can be tens of microamps while delivering 100 mA.
2. **Bad news:** "no DC current" is not "no current". The gate is a *capacitor*, and to *change* the gate voltage you must move charge into or out of it. The pass transistor's gate capacitance is about 7 pF — enormous, because the device is 2.5 mm wide. **This single fact is the reason your schematic contains a slew-rate enhancement block.**

```
C_gate ≈ W · L · C_ox = 2483 µm × 0.35 µm × 8.42 fF/µm² = 7.3 pF

Time to swing that gate by 1.4 V with only the amplifier's 2 µA:
  t = C·ΔV / I = (11.2 pF × 1.4 V) / 2 µA = 7.9 µs   ← far too slow
With the slew booster supplying 300 µA:
  t = (11.2 pF × 1.4 V) / 300 µA = 52 ns          ← 150× faster
```

$R_{DS(on)}$, power dissipation and heat

When a MOSFET is hard on and in triode it behaves as a resistor. That resistance is $\mathbf{R_{DS(on)}}$:

```
R_DS(on) ≈ 1 / [ µ_p C_ox · (W/L) · (V_SG − |V_TH|) ]

For M_P at full drive: = 1 / [71.2 µA/V² × 7093 × (1.6 − 0.51)]
                       = 1.82 Ω

Check against the spec:  V_drop = I·R = 0.1 A × 1.82 Ω = 182 mV  ✓ under the 200 mV budget
Heat in the pass device:  P = I²R = 0.1² × 1.82 = 18.2 mW
```

On a board that 18 mW would be a heatsinking question. On-chip it is a *layout* question: 18 mW concentrated in one 2.5 mm-wide device raises its local temperature, V_{TH} falls with temperature (about −1 mV/°C), and the device passes slightly more current — mild positive feedback called **thermal runaway** in power devices. At 18 mW it is harmless. It is the reason the pass device is laid out as hundreds of parallel fingers spread over an area rather than one long strip, and the reason the temperature-sensitive matched pairs (M_{N1}/M_{N2}) are placed far away from it.

The body diode — why it exists

Look at the cross-section again. The n+ source island sits inside p-type body. **An n-region touching a p-region is a PN junction — a diode.** You did not add it; it is an unavoidable consequence of building the transistor at all. Every MOSFET has two of them (source-body and drain-body).

```
        D
        │
        ├─┐
  G ─┤  ▲│   ← body diode: anode = body(p), cathode = drain/source(n) for NMOS
        ├─┘      (arrow points from p to n, i.e. from body toward drain)
        │
        S ── B    ← source and body tied together (standard practice)
```

Tying source to body shorts out one of the two diodes, leaving one from body to drain. **It must stay reverse-biased at all times.** For an NMOS with body at GND that means the drain must never go below about −0.6 V; for a PMOS with body at VIN, the drain must never go above VIN + 0.6 V. If it does, the diode conducts, current flows into the substrate, and in CMOS that can trigger **latch-up** — a parasitic thyristor turns on and shorts VIN to GND until the power is removed or the chip dies.

> IN YOUR CIRCUIT
>
> All PMOS bodies tie to VIN (1.8 V), all NMOS bodies to GND (0 V). No node ever leaves the 0–1.8 V window in normal operation, so every body diode is reverse-biased and invisible. It becomes visible only in two situations worth naming in a viva: **(a)** if V_{OUT} is pulled *above* V_{IN} by an external source, M_P's drain-body diode conducts backwards from output to input — the classic "reverse current" failure of a PMOS-pass LDO, and a reason real parts add a back-to-back blocking device; **(b)** during power-up before V_{IN} is established.

MOSFET as a switch vs as an amplifier

As a switch

Drive the gate all the way on or all the way off. In triode it is ~2 Ω; in cut-off it is essentially infinite. Used in digital logic and power switching. **Nothing in your circuit is used purely as a switch** — this is an analog circuit.

As an amplifier

Bias the gate *part way* on, in saturation. Now a small change in V_{GS} produces a proportional change in I_D. The constant of proportionality is the **transconductance $g_m = \partial I_D / \partial V_{GS}$**, measured in siemens (A/V). **Every device in your red box works this way.**

```
Three equivalent forms of g_m — use whichever the question gives you data for:
  g_m = µC_ox(W/L)·V_OV   =   2I_D/V_OV   =   √(2·µC_ox·(W/L)·I_D)   [strong inversion]
  g_m = I_D / (n·V_T)                                                [subthreshold — most efficient g_m per amp]

Small-signal output resistance (why gain is finite):
  r_o = 1 / (λ·I_D)      λ = channel-length modulation, ≈0.05 V⁻¹ at L=1 µm, ≈0.15 V⁻¹ at L=0.35 µm

Gain of one common-source stage:
  A_V = −g_m · r_o   ("intrinsic gain") — longer L gives more gain but is slower
```

Notice the trade-off buried in $r_o = 1/(\lambda I_D)$: **more bias current gives more g_m but less r_o.** This is why the adaptive bias block in your circuit raises the amplifier's *bandwidth* at heavy load without much changing its *DC gain* — a result we compute exactly in Part 13.

Correct ways to drive a MOSFET gate

On a board this means gate resistors and totem-pole drivers. On-chip, and specifically in your circuit, the rules are:

- **Never leave a gate floating.** An unconnected gate is a capacitor with nothing holding its charge. Leakage and capacitive coupling will drift it anywhere, including half-on, and a half-on 2.5 mm pass transistor is a short from VIN to GND. Every gate in your schematic is driven by something: a node, a mirror, or a bias.
- **Drive it from a low-impedance source when you need speed.** The amplifier output node driving V_G has a 10 MΩ output resistance — terrific for gain, terrible for speed into 11 pF. The slew booster exists precisely to provide a temporary low-impedance drive.
- **Respect the oxide.** $|V_{GS}|$ must never exceed the process limit (≈1.98 V for 1.8 V devices). In your circuit no gate can exceed VIN, so this is satisfied structurally — worth saying if asked about gate protection.

PART 04

When a MOSFET behaves like a diode

Your professor is pointing at something real and important: there are **two completely different mechanisms** by which a MOSFET can look like a diode, and eight devices in your schematic use one of them deliberately. Getting these two apart is a high-value viva answer.

First, what a diode is

A **diode** is a two-terminal device that conducts current in one direction (anode → cathode) and blocks it in the other. Forward-biased, a silicon PN diode drops roughly 0.6–0.7 V and then conducts exponentially. Reverse-biased, it passes only nanoamps — until it breaks down.

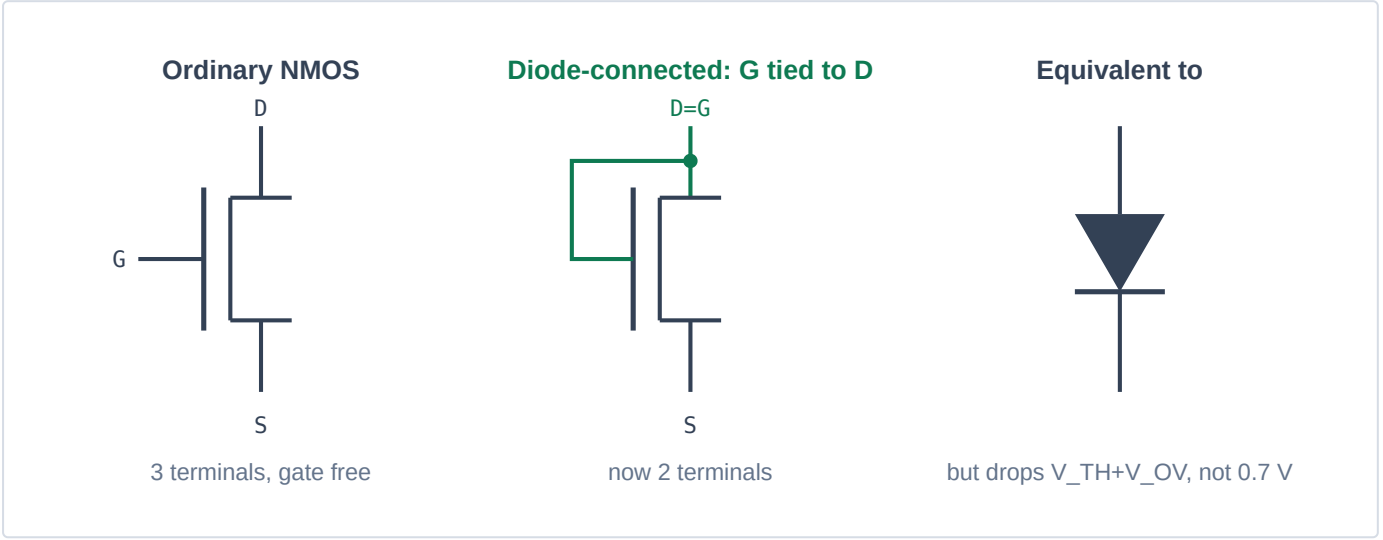
Mechanism 1 — the body diode (parasitic, unwanted)

Covered in Part 03: the source/drain islands form PN junctions with the body. This diode exists whether you want it or not, it is *parallel* to the channel, and it is *not controlled by the gate at all*. It conducts when the drain goes 0.6 V beyond the body.

It is the diode in a discrete power MOSFET datasheet that people use for freewheeling in motor drives. **In your circuit it is purely a hazard to be avoided, never a functional element.** If an examiner asks "does your circuit use the body diode?", the answer is no — and then you explain the reverse-current case from Part 03 to show you know why it matters anyway.

Mechanism 2 — the diode-CONNECTED MOSFET (deliberate, used 8 times in your schematic)

This is the one your professor means. It has nothing to do with the body diode. You take an ordinary transistor and **short its gate to its drain**.



Shorting gate to drain forces $V_{GS} = V_{DS}$. That single constraint turns a three-terminal amplifier into a two-terminal one-way element.

Why the short makes it diode-like

Follow the logic — this is the answer to give:

1. Tying G to D forces $\mathbf{V_{GS} = V_{DS}}$.
2. The saturation condition for an NMOS is $V_{DS} > V_{GS} - V_{TH}$. Substituting $V_{DS} = V_{GS}$ gives $V_{GS} > V_{GS} - V_{TH}$, i.e. $0 > -V_{TH}$, which is **always true**. So a diode-connected MOSFET is *always in saturation whenever it conducts at all* — it can never enter triode.
3. Therefore its current is set purely by the voltage across it: $I_D = \frac{1}{2}\mu C_{ox}(W/L)(V - V_{TH})^2$. That is a two-terminal device with a one-way, sharply rising I–V curve — a diode's defining behaviour.
4. Below $V < V_{TH}$ it is essentially off. Above, it conducts hard. The "knee" is at $\mathbf{V_{TH} \approx 0.5\ V}$, not 0.7 V.

Forward and reverse

```
FORWARD  (drain more positive than source, NMOS)
   ┌── V+ ──► drain ═══ channel ═══ source ──► V−
   Gate is held at drain potential, so V_GS = V, channel inverts, current flows.
   Voltage across the device settles at V_TH + V_OV ≈ 0.6–0.8 V for typical currents.

REVERSE  (you swap the applied polarity)
   Remember: the SOURCE of an NMOS is the LOWER terminal. Swapping polarity
   re-labels which island is the source. The gate is tied to the OTHER island,
   which is now the more negative one → V_GS becomes ≈ 0 → device is OFF.
   ► Blocks. Only the reverse-biased body diode's leakage flows (~pA).
   ► BUT: push more than ~0.6 V reverse and the BODY DIODE forward-biases and
     conducts. So the channel blocks but the parasitic junction does not.
```

> THE DISTINCTION THAT EARNS MARKS
>
> "The diode-connected MOSFET's forward conduction comes from the **channel**, controlled by the gate, with a knee at V_{TH}. Its reverse blocking comes from the channel turning off. The body diode is a *separate, parallel* junction that limits how much reverse voltage you can apply before the whole thing conducts anyway. They are not the same mechanism and the body diode plays no part in normal forward operation."

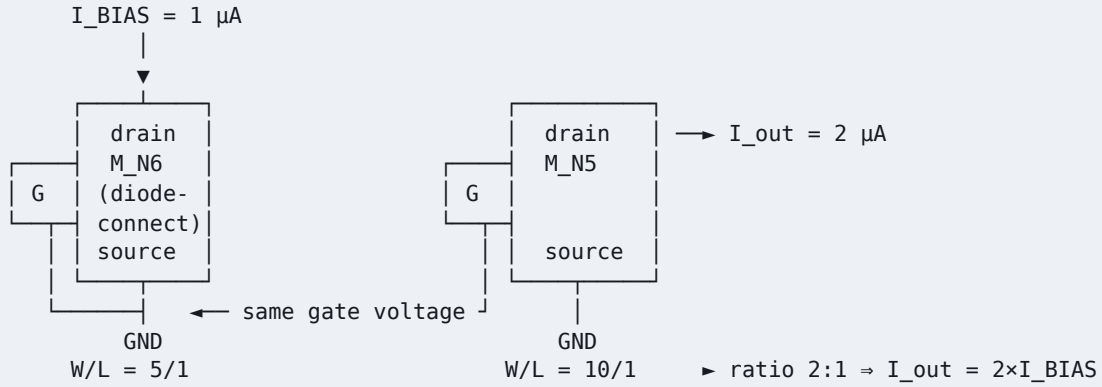
Where your schematic uses diode-connected devices — all eight of them

DEVICE	BLOCK	WHAT THE DIODE CONNECTION ACHIEVES
M_{N6}	Bias	**Current-to-voltage conversion.** I_{BIAS} is forced through it; it develops whatever V_{GS} is needed to pass exactly that current. That voltage is then copied to M_{N5}'s gate. This is the input half of a **current mirror**.
M_{P1}, M_{P2}	EA	**Low-impedance active load.** A diode-connected device presents resistance $1/g_m$ (tens of kΩ, not megohms), which keeps nodes X and Y fast and stable. They are also the input halves of the mirrors feeding M_{P5} and M_{P6}.
M_{N7}	EA	**Single-ended conversion.** Converts the left branch's current into a gate voltage that M_{N8} copies into the output node, so both halves of the differential signal end up at V_G.
M_{N10}	Adaptive bias	**Senses the replica load current and turns it into a bias voltage.** The replica current from M_{P9} has nowhere else to go, so M_{N10} develops exactly the $V_{GS} = V_{AB}$ that carries it. V_{AB} then drives M_{N9}. This is the *entire* adaptive-bias mechanism in one device.

DEVICE	BLOCK	WHAT THE DIODE CONNECTION ACHIEVES
M_{P7}, M_{P8}	Output	**A resistor made without a resistor.** Two in series from V_{OUT} to GND give a defined, area-efficient, weakly-conducting path. See Part 10.
M_{P12}–M_{P15}	SRE	**Pseudo-resistors.** A diode-connected PMOS biased at essentially zero current behaves as an enormous resistance — gigaohms. That is exactly what you need to set a DC gate bias without loading the AC signal. See Part 09.

Worked example: the simplest current mirror in your circuit

M_{N6} and M_{N5}. This pattern appears four times on the page; understand it once and you have understood the whole biasing scheme.



1. I_{BIAS} is forced into M_{N6}. Being diode-connected, M_{N6} has no choice: it settles at the gate voltage V_B that makes its drain current equal 1 µA.
2. M_{N5} sees the *same* V_{GS} (same gate wire, both sources at GND).
3. Same V_{GS}, same V_{TH}, same process ⇒ same current density. M_{N5}'s current is I_{BIAS} scaled by the W/L ratio.

Why this is better than a resistor: the output current barely depends on what voltage M_{N5}'s drain sits at (M_{N5} is in saturation, behaving as a current source), it occupies a fraction of the area, and it tracks I_{BIAS} over temperature because both devices change together. **That last point — matching — is why current mirrors are drawn as adjacent, identical devices and laid out interdigitated.**

When NOT to use a diode-connected MOSFET

- **When you need a real low-drop rectifier.** It drops $V_{TH} + V_{OV} \approx 0.7$ V, no better than a PN diode and far worse than a Schottky (0.3 V) or an active "ideal diode" controller.
- **When you need reverse blocking beyond 0.6 V.** The body diode takes over and conducts. This is why you cannot use one for reverse-polarity protection.
- **When you need a high-impedance node.** Its $1/g_m$ resistance *kills gain*. In your error amplifier, notice that M_{P1}/M_{P2} are diode-connected (low impedance, for speed and mirroring) but the *output* node V_G has no diode-connected device on it at all — M_{P6}, M_{N8} and M_{N3} all present r_o, giving 10 MΩ and therefore high gain. If you accidentally diode-connect M_{P6}, the loop gain collapses from 89 dB to about 20 dB and regulation is destroyed. That is a genuinely good "what if you changed this connection" answer.

- **When headroom is tight.** Each one eats V_{TH} + V_{OV} of your 1.8 V budget. Stacking three is usually impossible at this supply.

Diode vs diode-connected MOSFET — side by side

	PN JUNCTION DIODE	DIODE-CONNECTED MOSFET	MOSFET BODY DIODE
Forward drop	0.6–0.7 V, fixed by physics	V_{TH}+V_{OV}, **designable via W/L**	0.6–0.7 V
I–V law	exponential	square-law (or exponential if subthreshold)	exponential
Conduction is via	minority carriers across a junction	**the channel** — majority carriers	minority carriers
Gate control	none	self-controlled (G=D)	**none**
Reverse recovery	slow (stored charge)	**none** — majority carrier device	slow
Intended or parasitic	intended	**intended**	**parasitic**
In your schematic	none	8 devices, listed above	28 present, all kept off

PART 05

The complete netlist — every terminal of every device

If you memorise one page, make it this one. I read it off your schematic node by node. An examiner pointing anywhere on the page is pointing at a row in this table.

The nodes first

Every wire in a schematic is a **node** — a single electrical point, no matter how it is drawn. Your circuit has eleven that matter:

NODE	TYPICAL DC	IMPEDANCE	WHAT IT IS
VIN	1.8 V	low	Positive supply rail. Sources of every PMOS.
GND	0 V	low	Return rail. Sources of every NMOS.
V_{REF}	1.6 V	high-ish	Reference from an on-chip bandgap. Input to gates of M_{N1} and M_{N3}.
V_{OUT}	1.6 V	very low (closed loop)	The regulated output. Also the feedback signal — it drives M_{N2}, M_{N4} and the SRE directly.
V_G	0.2–1.7 V	**very high (≈10 MΩ)**	Gate of the pass device. The critical node in the whole circuit: high gain lives here, and so does 11 pF of capacitance.
X	≈1.2 V	low ($1/g_m$)	Left inner node of the EA: drains of M_{N1}, M_{P1}, M_{P3}; gates of M_{P1}, M_{P4}, M_{P5}.
Y	≈1.2 V	low ($1/g_m$)	Right inner node, mirror image: drains of M_{N2}, M_{P2}, M_{P4}; gates of M_{P2}, M_{P3}, M_{P6}.
OUT_L	≈0.6 V	medium ($1/g_m$ of M_{N7})	Left output branch: drains of M_{P5}, M_{N4}, M_{N7}; gates of M_{N7}, M_{N8}.
TAIL	≈0.25 V	medium	Common source node of all four input devices. Fed by M_{N5} (fixed) and M_{N9} (adaptive).
V_{AB}	0.4–0.7 V	low	Adaptive-bias voltage. Rises with load current. Drives M_{N9}'s gate.
A	self-biased ≈0.9 V	very high	Node between the two slew boosters. Drains of M_{P10} and M_{N11}; AC-couples into booster 2.

The netlist

DEVICE	TYPE	GATE	SOURCE	DRAIN	FUNCTION
M_{N6}	NMOS	self (=drain)	GND	I_{BIAS}	Diode-connected bias reference. Turns I_{BIAS} into a gate voltage.
M_{N5}	NMOS	from M_{N6}	GND	TAIL	Fixed tail current source for the input pairs (≈2 µA). Sets the light-load quiescent current.
M_{N9}	NMOS	**V_{AB}**	GND	TAIL	**Adaptive tail current.** In parallel with M_{N5}. Off at light load, adds ~20 µA at 100 mA. The bridge between the blue block and the red one.
M_{N1}	NMOS	V_{REF}	TAIL	X	Reference-side input device, mirrored path.
M_{N3}	NMOS	V_{REF}	TAIL	**V_G**	Reference-side input device, **direct path straight to the output node** — no mirror delay.
M_{N2}	NMOS	V_{OUT}	TAIL	Y	Feedback-side input device, mirrored path.

DEVICE	TYPE	GATE	SOURCE	DRAIN	FUNCTION
M_{N4}	NMOS	V_{OUT}	TAIL	OUT_L	Feedback-side input device, direct path into the left branch.
M_{P1}	PMOS	self (=X)	VIN	X	Diode-connected load at X; input side of the X−OUT_L mirror.
M_{P2}	PMOS	self (=Y)	VIN	Y	Same, mirror image, at Y.
M_{P3}	PMOS	**Y**	VIN	**X**	**Cross-coupled.** Adds positive feedback at X: partially cancels M_{P1}'s conductance and boosts gain.
M_{P4}	PMOS	**X**	VIN	**Y**	The other half of the cross-coupled pair.
M_{P5}	PMOS	X	VIN	OUT_L	Mirrors X's current into the left output branch. Gain factor B.
M_{P6}	PMOS	Y	VIN	**V_G**	Mirrors Y's current into V_G. **The pull-up half of the output stage.**
M_{N7}	NMOS	self (=OUT_L)	GND	OUT_L	Diode-connected; input side of the mirror that folds the left branch into V_G.
M_{N8}	NMOS	OUT_L	GND	**V_G**	Mirrors the left branch into V_G. **The pull-down half of the output stage.**
M_P	PMOS	V_G	VIN	V_{OUT}	**The pass transistor.** W/L ≈ 7100 (W ≈ 2.5 mm). Carries all 100 mA.
C_M	cap	between V_G and V_{OUT}		—	Miller compensation capacitor. Sets the dominant pole and feeds forward instantaneously on a transient.
M_{P8}	PMOS	self	V_{OUT}	to M_{P7}	Diode-connected bleeder, upper.
M_{P7}	PMOS	self	from M_{P8}	GND	Diode-connected bleeder, lower. Together they form a minimum-load / start-up path.
M_{P9}	PMOS	**V_G**	VIN	V_{AB}	**Replica of the pass device.** Same gate, same source ⇒ carries a scaled copy of I_{LOAD}. Ratio ≈ 1:5000.
M_{N10}	NMOS	self (=V_{AB})	GND	V_{AB}	Diode-connected. Converts the replica current into the bias voltage V_{AB}.
M_{P12}	PMOS	self	$BIAS_1$ ↔ gate of M_{P10}		Pseudo-resistor (GΩ). Sets M_{P10}'s DC gate voltage without shunting the AC signal.
M_{P14}	PMOS	self	$BIAS_2$ ↔ gate of M_{N11}		Same, for the NMOS half of booster 1.
M_{P10}	PMOS	AC-coupled from V_{OUT}	VIN	A	Booster 1, pull-up half.
M_{N11}	NMOS	AC-coupled from V_{OUT}	GND	A	Booster 1, pull-down half. Class-AB pair with M_{P10}.
M_{P13}, M_{P15}	PMOS	self	$BIAS_1$/$BIAS_2$ ↔ gates of M_{P11}/M_{N12}		Pseudo-resistors for booster 2.
M_{P11}	PMOS	AC-coupled from A	VIN	**V_G**	Booster 2, pull-up half. Drives the pass gate directly.
M_{N12}	NMOS	AC-coupled from A	GND	**V_G**	Booster 2, pull-down half. **This is the device that dumps current out of the pass gate on a load step.**

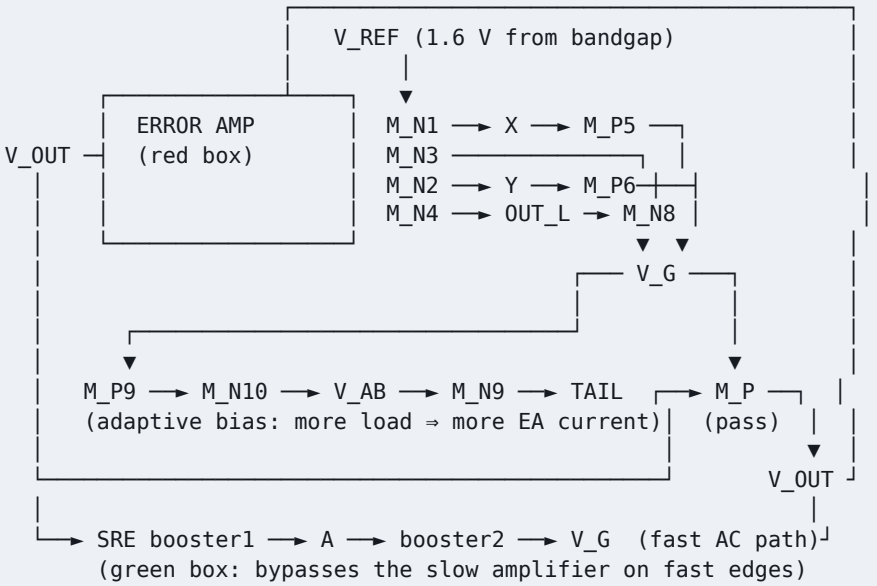
THREE THINGS TO NOTICE IN THAT TABLE

1. Four different devices drive the node $\mathbf{V_G}$: M_{P6} and M_{N8} (the amplifier's push–pull output), M_{N3} (the direct fast path), and M_{P11}/M_{N12} (the booster). Everything in this circuit converges on that one node.

2. $\mathbf{V_G}$ also leaves — it goes out to M_{P9}'s gate (adaptive bias) and into booster 2's output. The circuit senses its own control voltage to decide how much bias it needs.

3. $\mathbf{V_{OUT}}$ is fed back directly to M_{N2} and M_{N4} with no divider in between. That makes this a **unity-gain feedback** LDO: $V_{OUT} = V_{REF}$ exactly, with no division. It is the simplest, most accurate arrangement (no divider ratio error, no divider current) but it means **$\mathbf{V_{OUT}}$ can only ever equal $\mathbf{V_{REF}}$** — you cannot set 1.6 V from a 0.8 V bandgap. If your professor wants an adjustable output, this is the connection to change, and Part 17 shows how.

The signal flow, as one picture



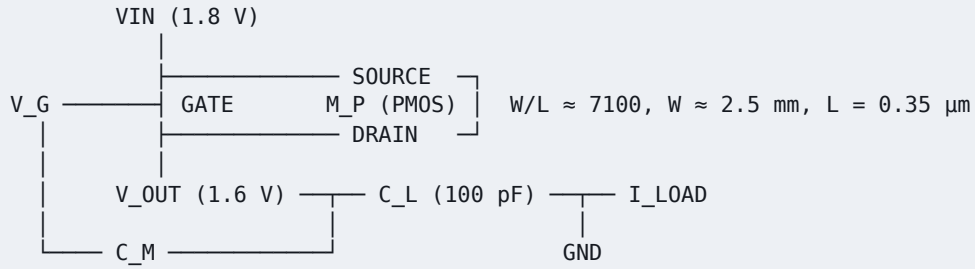
Three feedback loops share one node. The **main loop** ($V_{OUT} \to EA \to V_G \to M_P \to V_{OUT}$) does the regulating. The **SRE loop** ($V_{OUT} \to$ boosters $\to V_G \to M_P \to V_{OUT}$) is the same sign but faster and AC-only. The **adaptive-bias path** ($V_G \to M_{P9} \to V_{AB} \to M_{N9} \to$ TAIL) is not a signal loop at all — it modulates the main loop's *gain and bandwidth*. Saying that sentence cleanly is worth a lot.

PART 06

The pass transistor M_P and the compensation capacitor C_M

One transistor and one capacitor, but they decide the dropout, the efficiency, the stability and the transient response. Start here.

M_P — the connections and why each one



TERMINAL	GOES TO	WHY THERE	WHAT IF YOU MOVED IT
Source	VIN	A PMOS's source must be its most positive terminal. Tying it to the fixed rail means V_{SG} depends only on V_G, not on the output.	Swap source and drain and the "source" is now V_{OUT}: $V_{SG} = V_{OUT} - V_G$, so the device becomes a **source follower**. Gain drops from ~40 to below 1, dropout rises by a full V_{TH}, and the loop polarity inverts. The circuit does not regulate.
Gate	V_G	The control input. V_G low ⇒ large V_{SG} ⇒ device on hard. V_G near VIN ⇒ off.	Tie it to GND and the device is permanently fully on: $V_{OUT} = V_{IN} - I{\cdot}R_{DS(on)}$, unregulated. Tie it to VIN and there is no output at all. Leave it **floating** and it drifts to somewhere half-on — a slow short across the supply.
Drain	V_{OUT}	Where the regulated current is delivered.	This is also the feedback sense point; moving it breaks the loop.
Body	VIN (not drawn)	An n-well PMOS's body goes to the most positive node so both junctions stay reverse-biased.	Tie the body to V_{OUT} instead and the source–body junction forward-biases whenever $V_{IN} > V_{OUT} + 0.6$ V. Latch-up risk.

Why PMOS and not NMOS for the pass device?

This is the classic question and it has a clean quantitative answer. Compare the minimum dropout each can achieve:

PMOS pass (source on VIN, gate driven DOWN to turn on):
  $V_{DROPOUT,min} = V_{SD,sat} = I_{LOAD} \times R_{DS(on)}$ → can be made **as small as you like** by widening W
  With W/L = 7100: $V_{DROPOUT} = 0.1\ A \times 1.82\ \Omega = \mathbf{182\ mV}$ ✓

NMOS pass (source on V_OUT, gate must be driven ABOVE V_OUT):
  V_G must reach $V_{OUT} + V_{GS} = 1.6 + 0.5 + 0.15 = 2.25$ V — higher than VIN!
  → requires a **charge pump** to make a gate supply above VIN.
  Without one, $V_{DROPOUT,min} = V_{GS} \approx \mathbf{650\ mV}$ ✗

PMOS advantages (why yours uses it)

- Lowest possible dropout with no charge pump.
- Common-source connection gives voltage gain (~12–32 dB), which the loop needs.
- Naturally cuts off when V_G rises to VIN — safe default.

PMOS disadvantages (say these too)

- Hole mobility is ~5× lower ⇒ device is ~5× wider than an NMOS for the same current ⇒ ~5× the gate capacitance ⇒ the slew problem your SRE block exists to solve.
- Being a gain stage, it adds a pole at V_{OUT}, so the loop needs compensation.
- Its g_m and output pole both move by ~1000× across the load range — the hardest part of the stability problem.

The single most important non-obvious fact about M_P

g_m of the pass device is *not* constant. In strong inversion $g_m = \sqrt{(2 \cdot \mu C_{ox} \cdot (W/L) \cdot I_D)}$, so it scales with the **square root** of load current:

I_{LOAD}	V_{OV} OF M_P	REGION	G_{MP}	OUTPUT POLE $G_{MP}/2\pi C_L$
100 µA	20 mV	weak inversion	2.97 mS	4.7 MHz
1 mA	63 mV	moderate	29.7 mS	47 MHz
10 mA	199 mV	strong	100 mS	160 MHz
100 mA	629 mV	strong	318 mS	506 MHz

The output pole moves by a factor of 100 across the load range. That is the central stability problem of any LDO, and it is exactly what adaptive biasing is designed to track — see Part 13.

C_M — why a capacitor from V_G to V_{OUT}

C_M is connected *across* the pass transistor's gain stage — from its input (V_G) to its output (V_{OUT}). That specific placement is called **Miller compensation**, and it does three separate jobs:

1. **Pole splitting.** Seen from node V_G, a capacitor bridging a stage of gain $-A_2$ looks $(1 + A_2)$ times larger — the **Miller effect**. With $A_2 = 38$ and $C_M = 5$ pF, V_G sees 196 pF. That pushes the V_G pole down to ~80 Hz (the **dominant pole**) while simultaneously pushing the output pole *up*. Separating the two poles is what makes the loop stable.
2. **It sets the gain-bandwidth product.** GBW = $G_{m,EA}/(2\pi \cdot C_M)$. Choosing C_M *is* choosing the loop bandwidth.
3. **Instantaneous feed-forward.** This one is usually missed and is worth real marks. C_M is a direct capacitive path from V_{OUT} to V_G. When V_{OUT} suddenly dips by ΔV, charge conservation drops V_G immediately by $\Delta V \cdot C_M/(C_M+C_G) = 5/(5+6.2) = 0.45 \cdot \Delta V$ — *with no amplifier involvement and no delay at all*. That turns the pass device on harder within picoseconds, before any feedback loop has reacted.

THE PRICE OF C_M, AND HOW TO FIX IT

That same forward path creates a **right-half-plane (RHP) zero** at $\omega_z = g_{mp}/C_M$. An RHP zero adds gain but subtracts phase — the worst combination, and a classic cause of ringing. Standard fix: put a small resistor (or a MOSFET in triode acting as one) in series with C_M, sized $R_Z \geq 1/g_{mp}$, which moves the zero into the left half plane where it *helps* phase margin. Your schematic does not have one. **Adding a nulling resistor is one of the cleanest, most defensible improvements you can make to this design** — see Part 17.

What if you removed or moved C_M?

- **Removed:** the V_G pole (10 MΩ × 11 pF ≈ 1.4 kHz) and the output pole would sit close together at light load. Two nearby poles inside the loop bandwidth means < 45° phase margin: the output rings, and with a little more gain it oscillates outright. You would see a sine wave on V_{OUT} instead of a DC level.
- **Moved from V_G to GND instead of V_{OUT}:** you lose the Miller multiplication and the feed-forward. To get the same dominant pole you would need ~200 pF instead of 5 pF — 40× the silicon area. This is precisely why Miller compensation is used.
- **Made much larger:** more stable, but lower bandwidth ⇒ slower transient response ⇒ bigger output droop. This is the fundamental stability-vs-speed trade in the whole design.

PART 07

The error amplifier (red box) — all fifteen devices

This is the brain. It is a **cross-coupled current-mirror OTA with dual input pairs**. Long name, but each word is a design decision you can defend.

What it must do

Take two voltages (V_{REF} and V_{OUT}), subtract them, multiply the tiny difference by a huge number, and deliver the result as V_G. "Huge" matters: the DC accuracy of the output is set by the loop gain. A loop gain of 89 dB (28,000×) means a 1 V change in the gate needs only 35 µV of error at the input, so V_{OUT} sits within ~35 µV of V_{REF}. **Regulation accuracy is loop gain.**

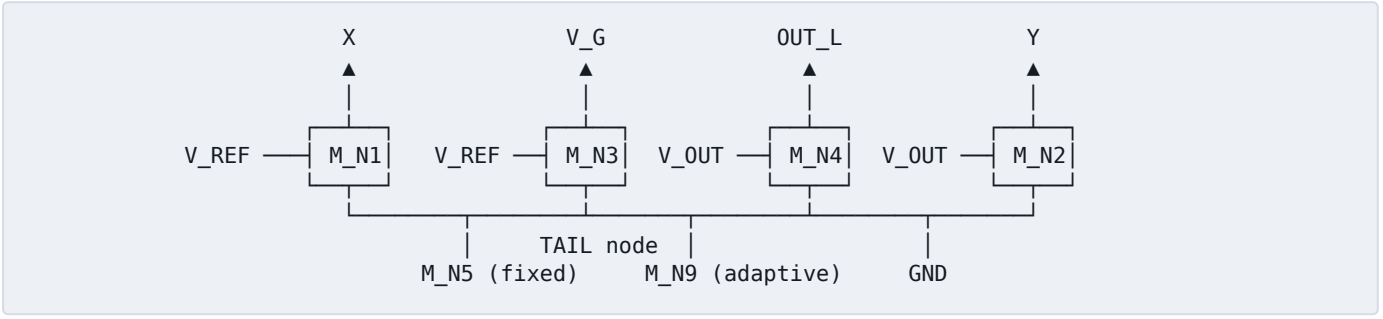
Layer 1 — the tail: M_{N6}, M_{N5}, M_{N9}

An amplifier needs a fixed, known current to work with. I_{BIAS} (1 µA, from a reference elsewhere) goes into diode-connected M_{N6}, which develops the gate voltage that carries it; M_{N5} copies it (scaled by W/L) into the TAIL node. M_{N9} sits in parallel and adds more current when V_{AB} tells it to.

- **Why a current source and not a resistor to ground?** A resistor's current would change with the tail voltage and with supply, so the amplifier's gain would move with V_{IN} — terrible power-supply rejection. A saturated MOSFET holds its current almost independent of the voltage across it.
- **What if M_{N5} is removed?** No tail current, no amplifier. V_G gets pulled to VIN by M_{P6}, M_P turns off, V_{OUT} = 0.
- **What if M_{N9} is removed?** The LDO still works, but only at the fixed light-load speed. Transient recovery at heavy load gets roughly 10× slower. This is the "what does the blue block actually buy you" question.

Layer 2 — the inputs: M_{N1}, M_{N3} (V_{REF}) and M_{N2}, M_{N4} (V_{OUT})

All four share the TAIL node. That makes them one **differential pair** in which each "side" is built from two transistors whose drains go to *different* places.



How a differential pair works: the tail supplies a fixed total current. The two sides compete for it. If V_{OUT} rises above V_{REF}, the V_{OUT} side (M_{N2}+M_{N4}) steals current from the V_{REF} side. The *total* never changes; only the *split* does, and that split is proportional to the input difference. That is how subtraction is done in analog.

WHY SPLIT THE INPUT INTO TWO PAIRS? — TRACE THE POLARITY YOURSELF

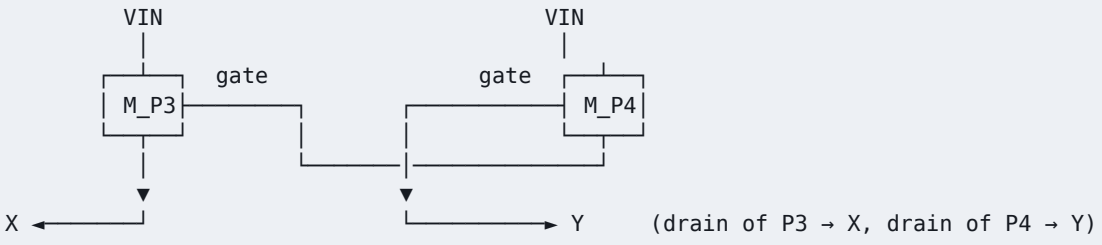
Suppose **V_{OUT} rises** (too high, loop must push it back down by turning M_P off, i.e. by raising V_G). Follow all four devices:

- M_{N2} conducts more ⇒ node Y's current rises ⇒ M_{P2}/M_{P6} mirror ⇒ M_{P6} pushes more current *into* V_G ⇒ **V_G rises** ✓
- M_{N4} conducts more ⇒ steals current from M_{N7} at OUT_L ⇒ M_{N8} mirrors less ⇒ pulls V_G down less ⇒ **V_G rises** ✓
- M_{N1} conducts less (tail is shared) ⇒ X falls ⇒ M_{P5} sources less into OUT_L ⇒ M_{N7}, then M_{N8}, carry less ⇒ **V_G rises** ✓
- M_{N3} conducts less ⇒ pulls down on V_G less directly ⇒ **V_G rises** ✓

All four paths agree. Nothing cancels. That is the design's whole point — and note that M_{N3} and M_{N4} reach the output nodes *directly*, in one device delay, while M_{N1}/M_{N2} must travel through a current mirror (two devices, plus the mirror's own pole). **The dual-pair structure buys transient speed and a push–pull output, not raw DC gain.** Be honest about that in the viva — claiming it "doubles the gain" is the mistake most students make, and is easy to disprove.

Layer 3 — the cross-coupled pair M_{P3}, M_{P4}

M_{P3}'s gate goes to node Y and its drain to node X; M_{P4} is the mirror image. Each one responds to the *other* node — that is **positive feedback**.



Why this raises the gain: the gain at node X is (current in) × (resistance at X). M_{P1}, being diode-connected, presents a low resistance $1/g_{m,MP1}$ that limits gain. M_{P3} injects current in the *opposite* sense — a **negative resistance** — which partially cancels M_{P1}. The net conductance shrinks, so the resistance and the gain grow. With $g_{m,MP3} = ½ g_{m,MP1}$, k = 2: the stage gain doubles for free, no extra current, no extra pole.

THE DANGER — AND THE DESIGN RULE

If $g_{m,MP3}$ ever **equals or exceeds** $g_{m,MP1}$, the net conductance goes to zero or negative and the circuit **latches**: nodes X and Y snap to opposite rails and stay there. The LDO is then dead until power-cycled. **Design rule: size $(W/L)_{MP3}$ at no more than about 0.5 × $(W/L)_{MP1}$, and verify across all process corners and temperatures**, because g_m ratios shift with corner. If an examiner asks "what is the risk in this circuit?", this is the best answer available. Simulate slow-NMOS/fast-PMOS at −40 °C and 125 °C and show it does not latch.

Layer 4 — the mirrors M_{P1}/M_{P5}, M_{P2}/M_{P6}, M_{N7}/M_{N8}

Three current mirrors, each doing the same trick from Part 04. The first two carry the differential signal up to the output branches with a gain factor $B = (W/L)_{MP5}/(W/L)_{MP1}$. The third, **$M_{N7}/M_{N8}$, is the differential-to-single-ended converter**: it takes the left branch's current and folds it into the right branch, so that both halves of the differential signal contribute to one output node V_G. Without it, half your signal would be thrown away and the amplifier would lose 6 dB and its ability to pull V_G down.

Layer 5 — the output node V_G

V_G is driven from above by M_{P6} and from below by M_{N8} and M_{N3}. Three devices in saturation, three r_o's in parallel, giving ≈10 MΩ — that high resistance is where the amplifier's voltage gain comes from, and it is also why the node is slow. **You cannot have both.** Everything else in the schematic is a way of buying speed back without giving up that resistance.

PART 08

Adaptive bias (blue box) — M_{P9}, M_{N10}, M_{N9}

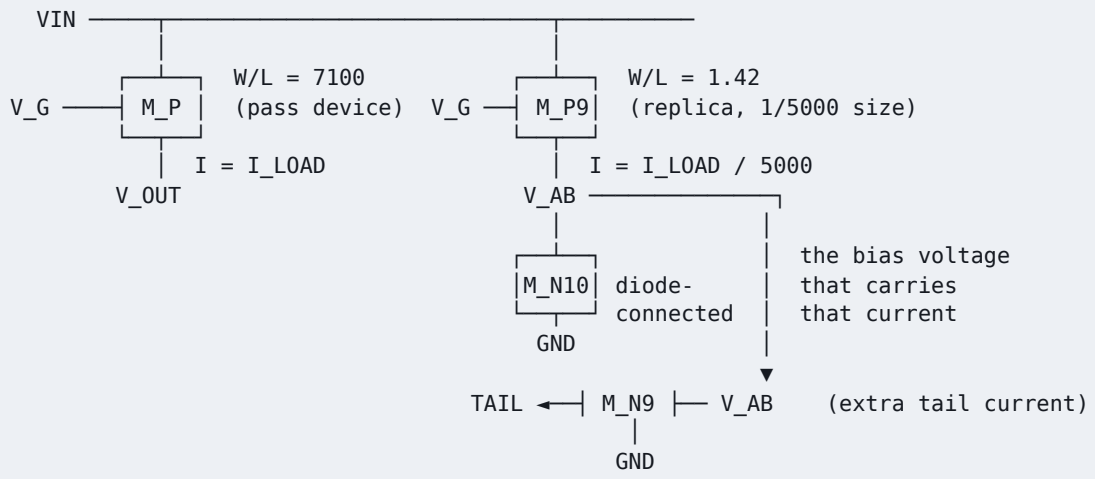
Three transistors that resolve the core conflict of low-power analog design. Small block, big idea.

The conflict

The amplifier's speed depends on its bias current. More current ⇒ more g_m ⇒ more bandwidth ⇒ faster transient recovery. But bias current is burned **all the time**, including when the chip is asleep and the load draws microamps. In a battery product that is the number that shows up in standby life.

```
Fixed at 2 µA:   great standby (η = 68 % at 100 µA load), but too slow at 100 mA
Fixed at 30 µA:  great transients at 100 mA, but η = 6 % at a 100 µA load
Adaptive:        2 µA at light load, 22 µA at heavy load  ← both
```

How the three devices do it



1. **M_{P9} is a scaled replica of the pass transistor.** Its gate is on the *same* node V_G and its source on the *same* rail VIN. Same V_{SG}, so its current density matches M_P's exactly, and its current is I_{LOAD} divided by the size ratio. **This is sensing the load current without a sense resistor** — no voltage lost, no power wasted. That is the elegant part.
2. **M_{N10} converts that current into a voltage.** Diode-connected, so it self-adjusts to whatever V_{GS} passes the replica current. Call it V_{AB}. Light load ⇒ tiny current ⇒ V_{AB} below threshold ⇒ M_{N9} is off. Heavy load ⇒ V_{AB} rises ⇒ M_{N9} turns on.
3. **M_{N9} injects that current into the amplifier's tail**, in parallel with the fixed M_{N5}. M_{N10} and M_{N9} together are simply another current mirror, with a programmable ratio you choose by sizing.

```
Choose the replica ratio. With (W/L)MP9 / (W/L)MP = 1/5000 and M_N10:M_N9 = 1:1 —
  I_LOAD = 100 µA  ⇒  I_MP9 = 20 nA   ⇒ M_N9 essentially off ⇒ I_tail = 2 µA
  I_LOAD =  10 mA  ⇒  I_MP9 = 2 µA    ⇒ I_tail = 2 + 2 = 4 µA
  I_LOAD = 100 mA  ⇒  I_MP9 = 20 µA   ⇒ I_tail = 2 + 20 = 22 µA

Quiescent current penalty at full load: 20 µA on 100 mA = 0.02 %  — free.
```

Why it makes the loop *more* stable, not less

This is the deep reason, and it is the strongest thing you can say about this block.

From Part 06, the non-dominant pole at the output is $f_{p2} = g_{mp}/(2\pi C_L)$, and $g_{mp} \propto \sqrt{I_{LOAD}}$. So the pole that threatens stability **moves to higher frequency** as the load increases. Meanwhile the loop's bandwidth is GBW = $G_{m,EA}/(2\pi C_M)$, and $G_{m,EA} \propto I_{tail}$.

```
Phase margin depends on the RATIO GBW : f_p2. Keep the ratio, keep the margin.

  Light load:   GBW = 2.4 MHz,  f_p2 =   4.7 MHz  ⇒ ratio 0.5  ⇒ PM ≈ 63°
  Heavy load:   GBW =  26 MHz,  f_p2 = 506 MHz    ⇒ ratio 0.05 ⇒ PM ≈ 87°

Without adaptive bias (tail fixed at 2 µA), heavy load would give
  GBW = 2.4 MHz against f_p2 = 506 MHz — stable, but 10× slower than it could be.
```

So: **adaptive biasing spends bias current exactly when the pass device's own g_m has opened up the stability headroom to use it.** Light load is the critical corner for stability; heavy load is the critical corner for speed; the block puts the current where each corner needs it.

> THE HONEST WEAKNESS — KNOW THIS BEFORE AN EXAMINER FINDS IT
>
> The sense signal is $\mathbf{V_G}$, which is the amplifier's own slow output. On a sudden load step, V_G hasn't moved yet, so M_{P9} hasn't sensed anything, so the extra bias arrives *after* the amplifier needed it. **Adaptive bias is a steady-state optimisation, not a transient one.** It makes the loop fast once it knows the load is heavy; it cannot help the very first nanoseconds of a load step. That is precisely the gap the slew-rate enhancement block fills — and being able to explain why you need *both* blocks, rather than one, is exactly the level of understanding a viva is testing.

What if you changed the connections?

- **M_{P9}'s gate tied to GND instead of V_G:** it becomes a fixed current source, not a replica. The adaptive behaviour disappears and you have simply raised the standby current permanently.
- **Replica ratio too large (say 1/500):** at 100 mA, M_{P9} carries 200 µA and the tail hits 202 µA. Now GBW at heavy load overtakes even the very high f_{p2}, and worse, the *mirror* poles inside the amplifier come into play. You trade the stability you were given back away. Also, standby current rises because the replica leaks.
- **Replica ratio too small:** M_{N9} never turns on meaningfully and the block does nothing — you have added three transistors and no benefit.

- **M_{N10} removed:** V_{AB} is floating. M_{P9}'s current has nowhere to go, so V_{AB} is pulled up towards VIN, M_{N9} turns on fully and dumps huge current into TAIL. The amplifier's bias explodes and the loop likely oscillates. **A diode-connected device is not optional — it is the thing that defines the node's voltage.**

PART 09

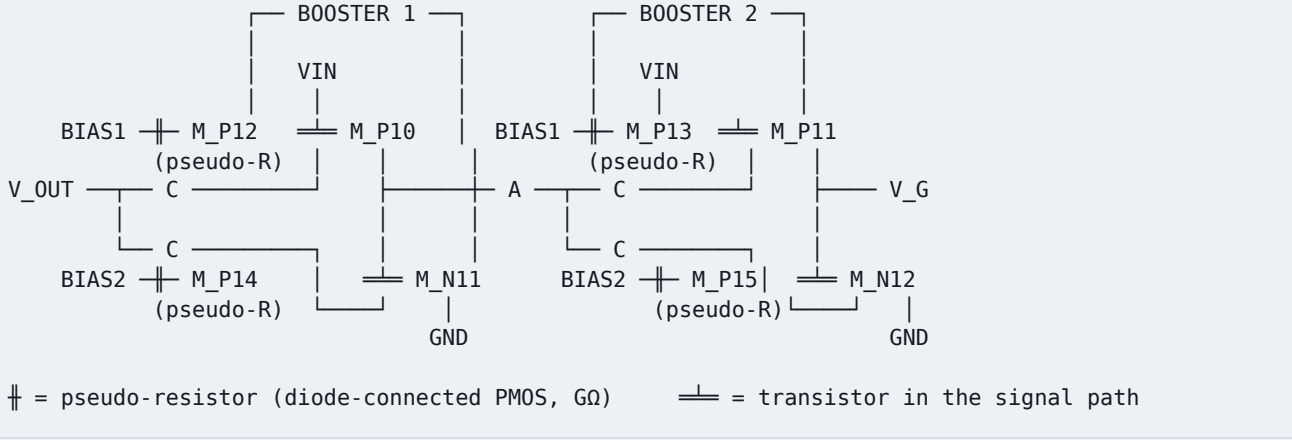
Slew-rate enhancement (green box) — eight devices, four capacitors

The block students find hardest to read, because two things happen at once: a signal path made of capacitors, and a bias path made of transistors pretending to be gigaohm resistors.

The problem it solves, in one calculation

> C at the V_G node = $C_{gate}(M_P)$ + C_M = 6.2 pF + 5 pF = 11.2 pF
> Current the amplifier can supply at light load ≈ I_{tail} = 2 µA
> Swing needed to go from "off" to "100 mA" ≈ 1.4 V
>
> t = C·ΔV/I = (11.2 pF × 1.4 V) / 2 µA = **7.9 µs**
>
> Meanwhile, with only 100 pF at the output, a 100 mA deficit drains the output at dV/dt = I/C = 0.1 A / 100 pF = 1000 V/µs.
>
> The output would be at zero volts long before the gate finished moving. **This is not a small inefficiency — without the booster the circuit fails outright.**

Reading the block: it is two identical stages in a chain



Each booster is a **capacitively-coupled class-AB inverting amplifier**. Unpack that:

- **Class-AB / push–pull:** a PMOS (pull-up) and an NMOS (pull-down) share the same output node and are driven by the *same* input. Whichever direction the input moves, one device turns on hard and the other turns off. It can source *and* sink large currents.
- **Inverting:** both are common-source stages. Input goes down ⇒ output goes up.
- **Capacitively-coupled:** the input reaches each gate through a capacitor, so only *changes* get through. In steady state the capacitors block, and the gates sit at whatever DC the pseudo-resistors set. That is why this block burns nanoamps at rest.

The pseudo-resistors M_{P12}–M_{P15}

Each gate needs a defined DC voltage or it floats (Part 03's cardinal sin). But connecting a normal resistor from BIAS to the gate would also short the AC signal to BIAS and kill the booster. You need something that is **a short at DC and an open at signal frequencies** — i.e. an enormous resistance, gigaohms, so that R·C gives a high-pass corner well below your signal but above DC.

A 1 GΩ polysilicon resistor in this process would be roughly a millimetre of serpentine. A diode-connected PMOS biased at essentially zero current does the same job in a few square micrometres: at V_{GS} just at or below threshold, its incremental resistance is in the gigaohm range. That is why M_{P12}–M_{P15} are drawn diode-connected between a BIAS node and a gate — **they are not amplifying anything, they are resistors made of silicon you already have.**

```
High-pass corner of each coupling path (this is the design equation):
  f_c = 1/(2π·R_pseudo·C_couple)
     = 1/(2π × 1 GΩ × 200 fF) = 796 Hz

Signals faster than ~800 Hz pass to the gate; anything slower is ignored. A load
transient has microsecond edges (MHz content) ⇒ passes fully. DC drift and the
regulated 1.6 V level ⇒ blocked. Exactly the wanted behaviour.
```

$BIAS_1$ sits near VIN, so each PMOS (M_{P10}, M_{P11}) rests just barely on. **$BIAS_2$** sits near GND, so each NMOS (M_{N11}, M_{N12}) also rests just barely on. Both devices in each booster idle in deep subthreshold at a few nanoamps — the "AB" in class-AB — so the block is ready to conduct instantly without waiting to turn on, yet costs almost nothing at rest.

Trace a load step through it — get the polarity right

```
Load suddenly demands 100 mA.  V_OUT begins to DROP.
     │
     ▼  (through coupling caps, instantly)
Gates of M_P10 and M_N11 both go DOWN by ≈ΔV
     │      M_P10 (PMOS): V_SG increases  ⇒ turns ON  hard   ── pulls A UP
     │      M_N11 (NMOS): V_GS decreases  ⇒ turns OFF        ── stops holding A down
     ▼
Node A goes UP  (stage 1 inverted the signal)
     │
     ▼  (through coupling caps, instantly)
Gates of M_P11 and M_N12 both go UP
     │      M_P11 (PMOS): V_SG decreases  ⇒ turns OFF
     │      M_N12 (NMOS): V_GS increases  ⇒ turns ON hard    ── SINKS current from V_G
     ▼
V_G is yanked DOWN ──────► M_P's V_SG increases ──────► M_P delivers more current
     │
     ▼
V_OUT recovers.  As it recovers, the coupling caps see no more change,
the boosters relax back to their nanoamp rest state, and the slow main
loop takes over holding the final DC value.
```

WHY TWO STAGES AND NOT ONE?

Polarity. Each common-source stage inverts. V_{OUT} falling must make V_G fall (to turn the PMOS pass device on). Falling → falling requires an *even* number of inversions. One stage would give V_{OUT} down ⇒ V_G up ⇒ pass device turns further **off** ⇒ V_{OUT} falls more ⇒ **runaway positive feedback**. Two stages fix the sign. They also give the gain needed to turn a 50 mV output dip into a volt-scale gate swing.

This is the highest-value single question in the green box. If asked "why two boosters", the answer is *sign, then gain* — in that order.

```
What the booster buys, numerically:
  Amplifier alone, light load:  SR = 2 µA / 11.2 pF   = 0.18 V/µs → 1.4 V in 7854 ns
  With adaptive bias engaged:   SR = 22 µA / 11.2 pF  =  1.96 V/µs → 1.4 V in 714 ns
  With SRE booster (300 µA):    SR = 300 µA / 11.2 pF = 26.7 V/µs → 1.4 V in 52 ns

                                                      150× faster than the amplifier alone.
```

Node A — the weak point worth naming

Node A carries the drains of M_{P10} and M_{N11} and nothing else but capacitors. **There is no DC path to set its voltage.** It self-biases at whatever level makes $|I_{MP10}| = I_{MN11}$, both in deep subthreshold at a few nanoamps. That equilibrium is:

- **Legitimate** — a self-biased inverter-like stage is a standard low-power technique.
- **Sensitive** — subthreshold currents vary exponentially with V_{TH} and temperature, so node A's resting level moves across process corners. In one corner it can sit near a rail, leaving the booster with no headroom in one direction.

What to do about it (and say in the viva): sweep the corners in simulation, plot node A's DC level across temperature and corner, and if it drifts too far, add a very weak pseudo-resistor from node A to a mid-rail bias, or replicate M_{P10}/M_{N11} as a diode-connected pair to define A. Raising a real weakness and then offering the fix is exactly how to answer "what are the limitations of your design?"

What if the coupling capacitors were removed, or replaced by wires?

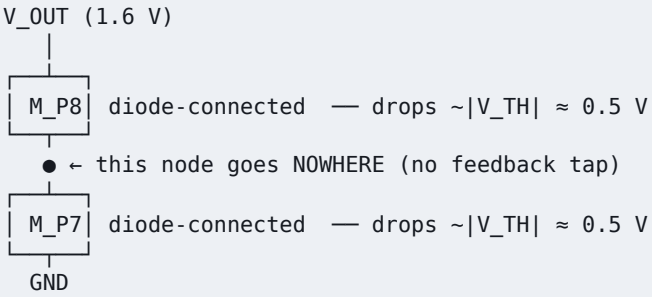
- **Replaced by wires (DC-coupled):** the boosters would see the DC level of V_{OUT} (1.6 V) on their gates. M_{N11} would be fully on, M_{P10} fully off, and booster 1 would sit hard against a rail drawing a large static current — destroying the quiescent current spec and doing nothing useful.
- **Removed (gates left with only the pseudo-resistor):** the boosters never see the transient. The block draws almost no current and provides no benefit. Behaviour reverts to the 7.9 µs slew of the amplifier alone.
- **Made much larger:** more signal reaches the gates (stronger boost), but the caps load V_{OUT} and node A, adding poles inside a feedback path. Too large and the SRE loop itself rings. Typical values are 100–500 fF for exactly this reason.

PART 10

M_{P7} and M_{P8} — the two devices everyone misreads

Small, easy to skip, and a favourite target because most students guess wrong about them.

Two diode-connected PMOS devices in series from V_{OUT} down to GND. There is **no wire leaving the node between them** — check your schematic and confirm this yourself, because it settles what they are.



What they are NOT

They are not a feedback divider. In a conventional LDO a resistive divider from V_{OUT} feeds a tapped fraction to the amplifier so that $V_{OUT} = V_{REF}(1 + R_1/R_2)$. Here the amplifier's input M_{N2}/M_{N4} takes **V_{OUT} directly**. There is no tap, so there is no division: this is a **unity-gain-feedback LDO with $V_{OUT} = V_{REF} = 1.6$ V.**

What they are

A **minimum-load (bleeder / pre-load) path**, built from transistors because a resistor of the same value would be far larger. They do three jobs:

1. **Guarantee a minimum load.** At truly zero load current the pass device operates at essentially zero current, its g_m collapses, the output pole falls to very low frequency and the loop's phase margin degrades badly. A small guaranteed bleed keeps the loop in a defined operating point. This is why commercial LDO datasheets often specify a minimum load.
2. **Discharge the output when the LDO is disabled or the load is removed.** Without a path to ground, V_{OUT} would sit charged on C_L indefinitely, and an overshoot would have no way to decay.
3. **Define a start-up path.** During power-up they give the output node a defined DC state instead of leaving it floating.

```
Bleed current — two series diode-connected PMOS across 1.6 V, so ~0.8 V each:
  each device sits at |V_GS| ≈ 0.8 V, i.e. V_OV ≈ 0.29 V, square-law:
  I = ½ · µ_p C_ox · (W/L) · V_OV²
    = ½ × 71.2 µA/V² × (W/L) × 0.29²  =  2.99 µA × (W/L)

  For a 3 µA bleed:  (W/L) ≈ 1, e.g. W = 0.5 µm, L = 0.5 µm — two tiny devices.

Quiescent cost: 3 µA. Against a 100 mA load that is 0.003 %; against a 100 µA
standby load it is a real 3 % — which is why the sizing matters and why some
designs switch this path off in standby.
```

> CHECK THIS WITH YOUR PROFESSOR
>
> If in the version you are given there *is* a wire leaving the node between M_{P8} and M_{P7} going to M_{N2}'s gate, then they are a **diode-connected feedback divider** instead, the topology becomes $V_{OUT} = 2 \times V_{REF}$ (approximately — the ratio of two diode drops is not precise), and V_{REF} would be 0.8 V. Both readings are defensible; which one applies is settled by that single wire. In the schematic as drawn, there is no such wire, so the bleeder reading is correct. **Saying "these two devices are a minimum-load bleeder, but had the node between them been tapped they would be the feedback divider" demonstrates you actually read the schematic rather than pattern-matched it.**

If you removed them

The LDO would still regulate. You would gain ~3 µA of standby current and lose: defined behaviour at zero load, output discharge, and some phase margin at the lightest loads. In simulation you would see a slow, poorly damped recovery from a load release (overshoot that takes a long time to come back down), because nothing is pulling the output down.

PART 11

Complete path trace: supply → load → ground

The "walk me through your circuit" question. There are two journeys to trace: the power path (where the amps go) and the control path (where the information goes).

Journey 1 — the power path

```
STAGE 1   VIN pad                       1.800 V    100 mA enters
  │       Purpose: supply rail from battery/buck
  │       Electrically: voltage may be noisy, sagging; current is the full load
  ▼
STAGE 2   M_P source → channel → drain   1.800 → 1.618 V    100 mA
  │       Purpose: the controlled series element
  │       Electrically: 182 mV is DROPPED here as heat (18.2 mW).
  │                     Device is in TRIODE, behaving as a 1.82 Ω resistor.
  │                     Its resistance is continuously adjusted by V_G.
  ▼
STAGE 3   V_OUT node                     1.600 V    100 mA
  │       Purpose: the regulated rail; junction of pass device, C_L,
  │                bleeder, load, and the feedback sense wire
  │       Electrically: voltage is now clean and load-independent.
  │       Current splits:   99.99 mA → load
  │                             3 µA → bleeder M_P8/M_P7
  │                             0 µA → C_L  (capacitors pass no DC)
  │                             0 µA → feedback (gates draw no current)
  ▼
STAGE 4   The load                       1.600 → 0 V    99.99 mA
  │       Purpose: does the useful work
  │       Electrically: consumes 160 mW
  ▼
STAGE 5   GND rail → GND pad             0.000 V    100 mA returns
          Purpose: the return path; must be low-resistance metal
          Electrically: any resistance here creates ground bounce that
                        appears directly as an output error.
```

Notice how little of the circuit is in this path: **one transistor**. Everything else exists only to decide how open that one transistor should be.

Journey 2 — the control path, during a real load step

The load jumps from 1 mA to 100 mA. Follow it in time order; each row is what an examiner wants to hear.

T	WHAT HAPPENS	WHERE IN THE SCHEMATIC
0	Load demands 99 mA more than M_P is supplying. The deficit comes out of C_L.	V_{OUT} node
0+	V_{OUT} starts falling at $I/C = 99\ mA/100\ pF \approx 990\ V/\mu s$.	C_L
~0 (instant)	**C_M couples the dip straight to the gate.** V_G drops by $0.45 \times \Delta V$ with zero delay. M_P immediately conducts more. No transistor had to "decide" anything.	C_M
<1 ns	The dip also passes through the SRE coupling caps onto four gates. M_{P10} turns on, M_{N11} off ⇒ node A rises ⇒ M_{N12} turns on hard.	Boosters 1 – 2

T	WHAT HAPPENS	WHERE IN THE SCHEMATIC
1–50 ns	**M_{N12} sinks ~300 µA out of the gate node**, slewing V_G down at ~27 V/µs. M_P opens up towards 100 mA. V_{OUT} stops falling and turns around. This is the undershoot minimum.	$M_{N12} \rightarrow V_G \rightarrow M_P$
50–200 ns	V_G is now low, so the replica M_{P9} conducts ~20 µA, M_{N10} raises V_{AB}, M_{N9} turns on, and the amplifier's tail current rises 2 µA → 22 µA. The main loop is now 10× faster than it was a moment ago.	$M_{P9} \rightarrow M_{N10} \rightarrow M_{N9}$
0.2–2 µs	With V_{OUT} no longer changing fast, the SRE coupling caps stop conducting and the boosters relax to nanoamps. The main loop takes over and settles the DC level precisely: M_{N2}/M_{N4} compare V_{OUT} to V_{REF} and trim V_G until the error is ~35 µV.	Error amplifier
steady	V_{OUT} = 1.6 V. Tail current stays at 22 µA while the load is heavy. Quiescent current is 25–30 µA.	whole loop

> THE SENTENCE THAT TIES THE WHOLE CIRCUIT TOGETHER
>
> "Three mechanisms respond on three different timescales: C_M feeds forward *instantaneously*, the slew booster acts in *nanoseconds* with large current but no DC accuracy, and the error amplifier acts in *microseconds* with poor speed but 89 dB of accuracy. Adaptive biasing then moves the amplifier's speed up to match the load. No single block could provide all four properties at once."

What happens on a load release (the other direction)

Often forgotten, and asked precisely because of that. Load drops 100 mA → 1 mA. Now M_P is delivering far too much current and V_{OUT} **overshoots upward**. The loop must raise V_G to shut the pass device off. Push–pull matters here: M_{P6} pulls V_G up, and on the SRE side the polarities all reverse (M_{N11} on, node A down, M_{P11} on, V_G yanked up). **Overshoot is usually worse than undershoot in a capless LDO**, because the only things that can remove the excess charge from the output node are the load itself and the tiny bleeder — and at 1 mA there is very little to absorb it. That is the third job of M_{P7}/M_{P8} from Part 10.

PART 12

Component-by-component reference table

Everything in one place, with the parameter that matters and the consequence of deletion.

COMPONENT	WHAT IT IS	WHY IT IS USED	KEY PARAMETER	IF REMOVED
M_P	Very wide PMOS, W≈2.5 mm	The controlled series element; all load current flows through it	W/L ≈ 7100; $R_{DS(on)}$ = 1.82 Ω; C_{gate} = 6–7 pF	No output at all. It *is* the regulator.
C_M	Miller compensation capacitor	Splits the poles, sets GBW, and feeds the output dip forward to the gate instantly	5 pF ⇒ GBW = 2.4 MHz at light load	Two poles collide near the unity-gain frequency ⇒ ringing or sustained oscillation.
C_L	100 pF of on-chip parasitic + intentional decoupling	Holds the output up during the loop's response delay	Sets the output pole with g_{mp}; sets the droop for a given step	Any load step causes a catastrophic droop; loop stability worsens.
M_{N1}–M_{N4}	Four NMOS input devices sharing one tail	Subtract V_{OUT} from V_{REF}; two direct paths, two mirrored	Matching (ΔV_{TH} sets offset); $g_m = I_D/nV_T$ ≈ 15 µA/V each	Remove M_{N3}/M_{N4}: still works, loses the fast direct path and push–pull drive. Remove M_{N1}/M_{N2}: no mirrored path, gain falls badly.
M_{P1}, M_{P2}	Diode-connected PMOS loads	Low-impedance loads at X and Y; input halves of the output mirrors	$1/g_m$ ≈ 30 kΩ	Nodes X,Y undefined; mirrors have no reference; amplifier does not bias up.
M_{P3}, M_{P4}	Cross-coupled PMOS pair	Negative resistance that cancels part of M_{P1}/M_{P2} ⇒ gain boost with no extra current or pole	**Ratio g_{m3}/g_{m1} must stay < 1** (target ≈0.5) or the circuit latches	Gain drops ~6 dB; regulation accuracy worsens. Circuit still functions — this is the safest device to delete if you are chasing robustness.
M_{P5}, M_{P6}	PMOS mirror output devices	Carry the signal to the output branches; M_{P6} is the pull-up on V_G	Mirror ratio B (2 assumed); r_o ≈ 20 MΩ	V_G has no pull-up ⇒ pass device can never be turned off ⇒ no regulation upward.
M_{N7}, M_{N8}	NMOS current mirror	Differential-to-single-ended conversion; M_{N8} is the pull-down on V_G	1:1 ratio; r_o ≈ 40 MΩ	Half the signal is discarded (−6 dB) and V_G cannot be pulled down ⇒ loop fails in one direction.
M_{N5}	Fixed tail current source	Sets the amplifier's baseline bias and therefore standby current	≈2 µA	Amplifier dead. V_G floats to VIN, V_{OUT} = 0.
M_{N9}	Adaptive tail current source	Adds bias current in proportion to load	Adds ≈20 µA at I_{LOAD} = 100 mA	Works, but ~10× slower settling at heavy load.
M_{N6}	Diode-connected bias reference	Converts I_{BIAS} into the gate voltage all tail sources use	I_{BIAS} = 1 µA	No bias anywhere. Whole circuit dead.
M_{P9}	Scaled replica of the pass device	Senses load current without a sense resistor	Ratio ≈ 1:5000	Adaptive bias inoperative.
M_{N10}	Diode-connected NMOS	Converts the replica current to the voltage V_{AB}	Sets the M_{N10}:M_{N9} mirror ratio	V_{AB} floats up, M_{N9} turns full on, tail current runaway, likely oscillation.
M_{P10}, M_{N11}	Class-AB booster stage 1	Amplifies and *inverts* the output dip	Idles at a few nA; peak current sets the boost	No slew enhancement; recovery reverts to ~8 µs.

COMPONENT	WHAT IT IS	WHY IT IS USED	KEY PARAMETER	IF REMOVED
M_{P11}, M_{N12}	Class-AB booster stage 2	Second inversion (restores polarity) and drives V_G directly	Peak sink ≈300 µA ⇒ SR = 27 V/µs	Same as above; also, with only stage 1 the polarity would be *wrong* and the circuit would become unstable.
M_{P12}–M_{P15}	Pseudo-resistors (diode-connected PMOS at ~0 current)	Set the DC gate bias of the boosters without shunting the AC signal	R ≈ 1 GΩ ⇒ f_c ≈ 800 Hz with 200 fF	**Booster gates float.** They drift to an arbitrary level — possibly fully on, drawing large static current, or fully off, giving no boost. Non-optional.
4 × coupling caps	MOS or MIM capacitors, ~100–500 fF	Pass only fast changes to the booster gates; block DC	C with R_{pseudo} sets the high-pass corner	No signal reaches the boosters; block becomes dead silicon.
M_{P7}, M_{P8}	Diode-connected PMOS bleeder	Minimum load, output discharge, defined start-up state	≈3 µA at W/L ≈ 1	Poor damping at zero load; output cannot discharge; overshoot persists.
V_{REF}	External input from an on-chip bandgap	The accuracy standard the whole loop copies	1.6 V; its tempco and noise appear *directly* at V_{OUT}	No target to regulate to. Output undefined.
I_{BIAS}	External input from a current reference	Sets all quiescent currents	1 µA	Whole circuit unbiased and dead.

> A PATTERN WORTH NAMING OUT LOUD
>
> Scan the "if removed" column. Every entry that says "circuit dead" is a **diode-connected device or a bias source**. Every entry that says "still works, just worse" is a **performance enhancement**. That split — *what is structural vs what is optimisation* — is the clearest way to show you understand the design rather than having memorised it.

PART 13

Design calculations, step by step

Every number below is derived from the process parameters in the first table. Redo any of them on paper and you can defend the whole design.

13.0 Process constants (TSMC 0.18 µm, typical)

Gate oxide thickness	T_{OX} = 4.1 nm
Oxide capacitance	$C_{ox} = \varepsilon_{ox}/T_{OX} = (3.9 \times 8.854\text{e-}12)/4.1\text{e-}9 =$ **8.42 fF/µm²**
Electron mobility	μ_n = 416.5 cm²/V·s ⇒ $\mu_n C_{ox}$ = **351 µA/V²**
Hole mobility	μ_p = 84.5 cm²/V·s ⇒ $\mu_p C_{ox}$ = **71.2 µA/V²**
Thresholds	V_{THN} = 0.50 V, $\lvert V_{THP}\rvert$ = 0.51 V
Thermal voltage	$V_T = kT/q$ = 25.9 mV at 27 °C; subthreshold slope factor n ≈ 1.3
Nominal supply	1.8 V

These are low-field, long-channel values. Real short-channel devices show velocity saturation and mobility degradation, so **hand calculations typically under-predict the width you need by 20–50 %**. Say this when you present — it shows you know the limits of the method, and it is the honest reason your simulated sizes will differ.

13.1 Pass transistor sizing (the first real design decision)

The worst case is full load at minimum input: V_{SD} = 200 mV, which is small, so M_P is in **triode**. The amplifier can realistically pull V_G down to about 0.2 V (it cannot reach 0 V — M_{N8} needs V_{DSsat}).

Given: I_D = 100 mA, $V_{SG} = V_{IN} - V_G = 1.8 - 0.2 = 1.6$ V, V_{SD} = 0.2 V

Triode: $I_D = \mu_p C_{ox} \cdot (W/L) \cdot [\,(V_{SG} - \lvert V_{TH}\rvert) \cdot V_{SD} - V_{SD}^2/2\,]$

Bracket = $(1.6 - 0.51)(0.2) - (0.2)^2/2 = 0.218 - 0.020 = 0.198$ V²

$W/L = I_D / (\mu_p C_{ox} \times 0.198)$
$= 0.1 / (71.2\text{e-}6 \times 0.198) =$ **7093**

Choose L = 0.35 µm (a little above minimum, for better matching and V_{DS} robustness)
⇒ W = 7093 × 0.35 µm = **2483 µm ≈ 2.5 mm**

Laid out as ~800 parallel fingers of W = 3 µm each, in a common-centroid array.

What the result means physically: a transistor 2.5 millimetres wide on a chip whose smallest feature is 0.18 µm. It is by far the largest device on the die, and its area — roughly 0.05 mm² with routing — usually dominates the LDO's total area. Every design decision downstream (gate capacitance, slew rate, the entire green box) is a consequence of this one number.

13.2 Gate capacitance and why the SRE exists

$C_{gs} \approx W \cdot L \cdot C_{ox}$ = 2483 µm × 0.35 µm × 8.42 fF/µm² = **7.32 pF**
Effective gate-node capacitance (≈85 % of that, plus C_M):
C_{node} = 6.2 pF + 5 pF = **11.2 pF**

Slew rate at V_G = I_{drive} / C_{node}, and V_G must swing ≈1.4 V:
I = 2 µA (EA alone, light load) → SR = **0.18 V/µs** → t = 7854 ns
I = 22 µA (EA + adaptive bias) → SR = **1.96 V/µs** → t = 714 ns
I = 300 µA (SRE booster) → SR = **26.7 V/µs** → t = 52 ns

13.3 Transconductance of the pass device across the load range

First check which region it is in at each current:
$V_{OV} = \sqrt{2 \cdot I_D / (\mu_p C_{ox} \cdot W/L)}$

I_D = 100 µA: V_{OV} = √(2×100e-6 / 0.505) = **19.9 mV** → below ~100 mV ⇒ weak inversion
use $g_m = I_D/(nV_T)$ = 100e-6/0.0337 = **2.97 mS**

I_D = 100 mA: V_{OV} = √(2×0.1 / 0.505) = **629 mV** ⇒ strong inversion
use $g_m = 2I_D/V_{OV}$ = 0.2/0.629 = **318 mS**

g_m changes by 107× while the load changes by 1000× (the √I law).

13.4 Error amplifier transconductance and gain

Derivation, in the form to reproduce on a board. Let g = g_m of one input device, B = the PMOS mirror ratio (M_{P5}/M_{P1}), k = the cross-coupling gain boost. Summing the four signal paths from Part 07:

$G_{m,EA} = (1 + B \cdot k) \cdot g$ ← derived by adding the four paths at V_G

Light load, I_{tail} = 2 µA, so each of the four devices carries 0.5 µA (subthreshold):
g = 0.5 µA / (1.3 × 25.9 mV) = **14.9 µA/V**
With B = 2 and k = 2: $G_{m,EA}$ = (1+4) × 14.9 = **74.3 µA/V**

Output resistance at V_G = $r_{o,MP6} \parallel r_{o,MN8} \parallel r_{o,MN3}$, $r_o = 1/(\lambda \cdot I_D)$, λ = 0.05 V^{-1} (L = 1 µm)
$r_{o,MP6}$ = 1/(0.05 × 1 µA) = 20 MΩ
$r_{o,MN8}$ = $r_{o,MN3}$ = 1/(0.05 × 0.5 µA) = 40 MΩ
$R_{out,EA}$ = (1/20 + 1/40 + 1/40)$^{-1}$ MΩ = **10 MΩ**

A_1 = $G_{m,EA}$ × $R_{out,EA}$ = 74.3 µA/V × 10 MΩ = **743 (57.4 dB)**

A RESULT WORTH POINTING OUT UNPROMPTED

Repeat the calculation with the tail at 22 µA (heavy load): $G_{m,EA}$ rises 11× to 817 µA/V, but $R_{out,EA}$ *falls* 11× to 0.91 MΩ, because $r_o = 1/(\lambda I_D)$. **A_1 = 743 either way.** Adaptive biasing therefore buys *bandwidth* essentially for free without changing DC gain. (In reality the cancellation is not exact — λ varies with V_{DS} and the devices move from weak to moderate inversion — so simulation will show A_1 changing by a few dB. Say that too.)

13.5 Loop gain, line and load regulation

```
Second-stage gain:  A2 = gmp × (RL || ro,MP),  with λp = 0.15 V⁻¹ at L = 0.35 µm

At ILOAD = 100 µA: RL = 16 kΩ, ro = 67 kΩ, R2 = 12.9 kΩ
   A2 = 2.97 mS × 12.9 kΩ = 38.3
   Loop gain T = A1·A2 = 743 × 38.3 = 28 455  (89.1 dB)

At ILOAD = 100 mA: RL = 16 Ω, ro = 67 Ω, R2 = 12.9 Ω
   A2 = 318 mS × 12.9 Ω = 4.10
   Loop gain T = 743 × 4.10 = 3 045  (69.7 dB)

Load regulation = closed-loop output resistance = R2 / (1 + T)
   at 100 mA: 12.9 Ω / 3046 = 4.2 mΩ
   ⇒ ΔVOUT over the full 0→100 mA range ≈ 0.1 A × 4.2 mΩ = 0.42 mV  (0.026 %)

Static error referred to the input = (gate swing)/T = 1.4 V / 28455 = 49 µV
```

Interpretation: **the output moves less than half a millivolt as the load goes from nothing to full.** That is the payoff of 89 dB of loop gain, and it is the number to quote when asked "how good is your regulation?"

13.6 Stability — poles, GBW and phase margin

```
Dominant pole (Miller-multiplied, at V_G):
  f_p1 = 1 / [ 2π · R_out,EA · (1 + A_2) · C_M ]
       = 1 / [ 2π × 10 MΩ × 39.3 × 5 pF ] = 81 Hz   (light load)

Unity-gain frequency:
  GBW = G_m,EA / (2π·C_M) = 74.3e-6 / (2π × 5e-12) = 2.36 MHz   (light load)
                                                   = 26.0 MHz   (heavy load)

Non-dominant (output) pole:
  f_p2 = g_mp / (2π·C_L) = 2.97e-3 / (2π × 100e-12) = 4.73 MHz  (light load)
                                                    = 506 MHz   (heavy load)

Phase margin:  PM = 90° − arctan(GBW / f_p2)
  Light load: 90 − arctan(2.36/4.73)  = 63°   ✓
  Heavy load: 90 − arctan(26/506)     = 87°   ✓

RHP zero from C_M:  f_z = g_mp/(2π·C_M) = 2.97e-3/(2π×5e-12) = 94 MHz (light load)
  − comfortably above GBW here, but at very light loads it migrates down. Add a
    nulling resistor R_Z ≥ 1/g_mp ≈ 337 Ω in series with C_M to remove the risk.
```

READING THE RESULT

Light load is the critical corner. That is counter-intuitive and is often the whole point of an LDO stability question: at heavy load the pass device's g_{mp} pushes the output pole far away, but at light load g_{mp} collapses and the output pole drops right down to the loop's unity-gain frequency. A phase margin of 63° at light load is a good target (below 45° rings; above 75° is over-damped and slow).

Choosing C_M from scratch

```
Design rule: put GBW at or below half the worst-case non-dominant pole.
  Target GBW ≤ f_p2,min/2 = 4.73 MHz / 2 = 2.37 MHz
  C_M ≥ G_m,EA / (2π × 2.37 MHz) = 74.3e-6/(2π×2.37e6) = 4.99 pF  ⇒ choose 5 pF

Area cost: a 5 pF MOS capacitor ≈ 5 pF / 8.42 fF/µm² ≈ 594 µm²  (about 24 µm square)
```

13.7 Transient droop — the calculation that reveals a real design limit

Do this one carefully; it produces the most useful engineering insight in the whole project.

```
While the loop is catching up, the load's extra current comes out of C_L:
  ΔV_droop ≈ ΔI_LOAD × t_response / C_L
with  t_response ≈ 1/(2π·GBW) = 1/(2π × 26 MHz) = 6.1 ns  (heavy-load bandwidth)

For a 0 → 100 mA step:
  ΔV = 0.1 A × 6.1 ns / 100 pF = 6.1 V   ← impossible: the output would hit 0 V

Invert the formula to find what 100 pF can actually handle:
  ΔI_max = C_L · ΔV_allowed / t_response
         = 100 pF × 50 mV / 6.1 ns = 0.82 mA

Or find the capacitance a real 100 mA step needs:
  C_L,min = ΔI · t_response / ΔV = 0.1 × 6.1 ns / 0.1 V = 6.1 nF
```

SAY THIS IN YOUR VIVA BEFORE ANYONE ASKS

An output-capacitorless LDO with C_L = 100 pF cannot hold a fast 0–100 mA step within tens of millivolts. The physics forbids it, and no amount of slew enhancement changes the arithmetic. What the SRE and the adaptive bias genuinely do is minimise $t_{response}$, and that is the only lever available — but 100 pF is simply too small a reservoir for 100 mA.

What this means for your project, concretely:

- Either the 100 mA figure is a **DC rating** (the pass device is sized for it, which it is) while the transient spec applies to much smaller, or much slower, steps —
- or the design needs several nanofarads of on-chip decoupling, which is a large but real area cost —
- or the load step must have an edge time long compared with the loop's settling time, so the loop tracks it rather than being caught out.

Sanity-check the formula on a part you can look up: a commercial 300 mA LDO with a 1 µF output capacitor and ~1 MHz bandwidth gives ΔV = 0.3 × 159 ns / 1 µF = 48 mV, which matches published datasheet undershoot figures. The formula is sound; the 100 pF specification is what does not fit.

Presenting this as your own finding, with the arithmetic, is stronger work than reproducing a claimed 20 mV number you cannot derive.

13.8 Quiescent current, power and efficiency

I_Q budget:

I_{BIAS} branch (M_{N6})	1 µA
EA tail (M_{N5})	2 µA
EA mirror branches (M_{P5},M_{P6},M_{N7},M_{N8})	~3 µA
Bleeder M_{P7}/M_{P8}	3 µA
SRE at rest (8 devices, deep subthreshold)	<0.1 µA
Adaptive bias branch at light load	~0 µA

$I_{Q,light} \approx$ **9 µA** $I_{Q,heavy} \approx 9 + 20 =$ **29 µA**

Power in the pass device: $P = (V_{IN}-V_{OUT})\cdot I_{LOAD} = 0.2 \times 0.1 =$ **20 mW**

Power in the control circuitry: 1.8 V × 29 µA = **52 µW** (0.26 % of the total)

Efficiency $\eta = V_{OUT}I_{LOAD} / [V_{IN}(I_{LOAD}+I_Q)]$

- at 100 mA: (1.6×0.1)/(1.8×0.100029) = **88.9 %**
- at 100 µA: (1.6×100e-6)/(1.8×109e-6) = **81.5 %**
- at 10 µA: (1.6×10e-6)/(1.8×19e-6) = **46.8 %** ← I_Q now dominates

The last line is why low quiescent current matters and why the adaptive-bias block was worth three transistors: at deep standby, **the regulator's own consumption becomes the load**.

13.9 The on-chip equivalents of the board-level items in your brief

BOARD CONCEPT	ON-CHIP EQUIVALENT HERE	VALUE / STATUS
Pull-up / pull-down resistor	Current mirrors and pseudo-resistors	M_{P5}/M_{P6} pull up, M_{N7}/M_{N8} pull down, M_{P12}–$M_{P15} \approx 1$ GΩ
Gate resistor	Not needed — no wire inductance on-chip	The nulling resistor in series with C_M (≥337 Ω) is the only series resistance worth adding, and it is for the RHP zero, not for damping
Fuse / over-current protection	Current-limit loop (not present)	See Part 17 — this is your best single improvement
Heatsink	Layout: multi-finger device, thermal spreading, distance from matched pairs	18 mW is modest but locally concentrated
Bulk decoupling capacitor	C_L = 100 pF of on-chip MOS/MIM capacitance	Deliberately minimal — that is what "capacitorless" means, and Part 13.7 shows the price

PART 14

Why this design — advantages, limits and the alternatives it beat

Judgement questions. These separate a student who built the circuit from one who understands why it is shaped this way.

Advantages

- **No external capacitor** — saves a pin, a component, and board area. Lets you put many independent LDOs on one SoC, one per power domain.
- **Low quiescent current with good transient response**, which are normally mutually exclusive. This is the design's central claim.
- **Low dropout** (200 mV) from the PMOS pass device with no charge pump.
- **High DC accuracy** — 89 dB loop gain gives ~0.4 mV load regulation.
- **Unity-gain feedback** removes divider ratio error and divider quiescent current entirely.
- **Single-stage amplifier** — high gain achieved by cross-coupling rather than cascading, so there is one less pole to compensate.

Disadvantages and limits

- **The 100 pF transient limit of Part 13.7** — the honest ceiling on this topology.
- **Latch-up risk in the cross-coupled pair** if $g_{m,MP3} \geq g_{m,MP1}$ in any corner.
- **Node A in the SRE is self-biased**, so its operating point is corner-sensitive.
- **No current limit, no thermal shutdown, no UVLO, no soft-start** — not a production-ready part as drawn.
- **RHP zero from C_M is uncompensated** (no nulling resistor).
- **PSRR degrades at high frequency** because C_M couples VIN noise through the pass device with no cascode to block it.
- **Reverse current**: if V_{OUT} is forced above V_{IN}, M_P's body diode conducts backwards.
- **V_{OUT} is fixed at V_{REF}** — not adjustable without adding a divider.

Alternatives, and why they were not chosen

ALTERNATIVE	WHAT IT WOULD GIVE	WHY NOT HERE
Simple 5-transistor OTA + PMOS pass	Far simpler, easy to explain	Gain ≈ 40 dB, not 89 dB ⇒ ~30× worse load regulation; slew rate at V_G limited to microseconds; fails any transient spec without an external capacitor.
Two-stage cascaded amplifier	More gain by brute force	Adds a third pole inside the loop. With the output pole already sweeping 100× with load, three poles make compensation very hard at 1.8 V. The cross-coupled pair buys the same gain without the pole.
NMOS pass device (source follower)	Low output impedance, easy stability (the pole at V_{OUT} is well damped by the follower)	Needs V_G above VIN ⇒ a charge pump, which brings switching noise, area, and its own quiescent current. Dropout rises to ~650 mV without one. Genuinely the better choice at higher supplies — worth saying.
Resistive feedback divider	Adjustable output voltage	A divider drawing even 1 µA is 10 % of the light-load quiescent budget; the resistors would be physically large and poorly matched; and the divider adds a pole with the amplifier's input capacitance. Unity feedback avoids all three.

ALTERNATIVE	WHAT IT WOULD GIVE	WHY NOT HERE
Large external output capacitor (1 µF)	Transient response solved trivially; Part 13.7's problem disappears	Costs a pin and a discrete component per power domain. On an SoC with a dozen domains that is a dozen pins. The whole point of the capacitorless approach is to trade circuit complexity for pins and BOM.
Buck (switching) converter	>90 % efficiency even at large V_{IN}–V_{OUT}	Generates switching noise at its own clock frequency — precisely what a sensitive analog load is trying to escape. Needs an inductor. Real systems use a buck *then* an LDO.
Flipped Voltage Follower (FVF) LDO	Very fast, naturally low impedance, popular in recent capless work	Lower DC loop gain, so worse line/load regulation; and its output is set by a V_{GS} rather than a clean reference comparison. A legitimate alternative and a good answer if asked "what else could you have done?"

Failure points, ranked by likelihood

1. **Instability at light load** — the phase margin corner (Part 13.6). Symptom: ringing or oscillation on V_{OUT} with no load.
2. **Cross-coupled pair latching** in a fast-PMOS corner. Symptom: V_{OUT} stuck at 0 or at V_{IN}, unrecoverable without a power cycle.
3. **Node A drifting to a rail** across corners, disabling the SRE in one direction. Symptom: asymmetric transient response — fast undershoot recovery, slow overshoot recovery, or vice versa.
4. **Start-up hang.** At power-on every node is at 0 V, so $V_{AB} = 0$, M_{N9} off, and the amplifier has only M_{N5} — which needs M_{N6}'s bias, which needs I_{BIAS}, which may itself have a start-up condition. Symptom: output never comes up. Fix: a dedicated start-up circuit, and verify with a slow V_{IN} ramp in simulation.
5. **Over-current destruction** with no current limit. A short at the output puts V_{OUT} at 0 V, so V_{SD} across $M_P = 1.8$ V while the loop drives V_G fully down. The device is then in saturation delivering its maximum, and dissipating 1.8 V × several hundred mA — over a watt in a 2.5 mm device. It will be destroyed.

PART 15

What can go wrong — cause, symptom, detection, prevention

Translated into what these faults actually mean for an on-chip circuit, not a breadboard.

FAULT	CAUSE	WHAT HAPPENS	HOW TO DETECT	HOW TO PREVENT
Gate left floating	A missing connection in the netlist or a broken metal in layout — most likely on a booster gate, where the only DC connection is a gigaohm pseudo-resistor	The gate drifts on leakage to an arbitrary level. If it lands half-on, a wide device conducts continuously: large static current, possibly a VIN-to-GND short through M_P	DC operating-point simulation: look for a node with no DC path, and for unexpected branch currents. In silicon: anomalously high I_Q	Never leave a gate unconnected. Run a "floating gate" / ERC check. Verify each pseudo-resistor's DC continuity in simulation
Excessive gate voltage	V_{IN} above the 1.8 V process rating, or an ESD event on a pin	The 4.1 nm oxide punctures. Permanent, one-way. The gate becomes resistively shorted to the channel	Gate leakage current jumps from pA to µA; device no longer controllable	Add UVLO/OVLO; ESD diodes on every pad; never exceed $V_{IN,max}$ = 1.98 V
Output short to ground	A defect, or an external short	The loop sees V_{OUT} = 0, drives V_G fully down, M_P delivers its saturation current at V_{SD} = 1.8 V ⇒ >1 W locally ⇒ metal or device destruction within milliseconds	In simulation: sweep a load resistance to zero and watch I_{MP}. In silicon: supply current spike, then open-circuit	**Add a current limit** (Part 17). Also add thermal shutdown
Reverse output current	V_{OUT} forced above V_{IN} by another supply or a large inductive load	M_P's drain–body diode forward-biases. Current flows backwards from output to input regardless of V_G. Can trigger latch-up	Simulate with V_{OUT} forced to V_{IN}+0.5 V and observe the substrate/well current	Back-to-back pass devices, or a reverse-current comparator that disconnects. Note the limitation if you do not add one
Wrong device size on M_{P3}/M_{P4}	Sizing them equal to or larger than M_{P1}/M_{P2}	Positive feedback exceeds unity. Nodes X and Y latch to opposite rails. The amplifier stops amplifying; output sticks	Transient simulation across all corners and temperatures; look for X and Y splitting apart and staying split	Keep $(W/L)_{MP3} \le 0.5 \times (W/L)_{MP1}$ and verify the margin in the fast-PMOS/slow-NMOS corner at both temperature extremes
Insufficient supply	V_{IN} drops toward V_{OUT}	Dropout: loop saturates, V_{OUT} follows V_{IN}. Below ~1.1 V branches leave saturation and the loop can behave unpredictably on the way down	Slow V_{IN} ramp in simulation, plotting V_{OUT} and every branch current	UVLO that disables the LDO cleanly below a threshold, rather than letting it operate unbiased
Missing or resistive ground	Narrow ground metal carrying 100 mA	IR drop puts the load's ground above the amplifier's ground. The amplifier regulates to the wrong reference; the error appears 1:1 at the load	Post-layout simulation with extracted parasitics; or measure V_{OUT} at the load rather than at the pin	Wide power metal, separate analog and power grounds joined at one point, and Kelvin sensing of V_{OUT} at the load
Overheating	Sustained maximum load at maximum dropout	V_{TH} falls with temperature, so bias currents shift and V_{OUT} drifts. Sustained, it degrades reliability (electromigration in the pass device's metal)	Simulate at 125 °C; check current density in the pass device's metal against the process EM limit	Multi-finger layout, thick top-metal straps, thermal shutdown, and keep matched pairs away from the hot device

FAULT	CAUSE	WHAT HAPPENS	HOW TO DETECT	HOW TO PREVENT
Oscillation (light load)	Insufficient phase margin when g_{mp} collapses	V_{OUT} becomes a sine wave at roughly the unity-gain frequency instead of a DC level	AC/STB analysis — break the loop and plot gain and phase vs frequency, sweeping I_{LOAD} from 1 µA to 100 mA	Size C_M for the worst-case (lightest-load) pole; keep a minimum load via M_{P7}/M_{P8}; add the nulling resistor
Start-up hang	Every bias derives from another bias, and all start at 0 V	The circuit sits in a stable all-off state; output never rises	Transient simulation from $V_{IN} = 0$ with a realistic ramp (µs to ms), all node voltages plotted	A start-up circuit that injects current until the bias loop is alive, then switches itself off
Reversed source/drain on a PMOS	Schematic error — easy to make, since the two are physically identical	The device becomes a source follower or stays off entirely; gain collapses or the branch never biases	DC operating point: check every device's region field. Any device reported in "cutoff" or "subthreshold" that should be saturated is the clue	Always check the operating-point report device by device before trusting any other simulation

THE HABIT WORTH ADOPTING

Before any AC or transient run, print the **DC operating point** and check every one of the 28 devices: its region, its I_D, its V_{GS} and V_{DS}. Almost every fault in this table shows up there first, and being able to say "I verify the operating region of every device before I trust a simulation" is the single most professional thing you can say in a viva.

PART 16

Viva questions and answers

Answer in one or two sentences, then stop. Volunteering a third sentence is how students talk themselves into trouble.

A. Twenty fundamentals

1 What is a MOSFET?

A four-terminal voltage-controlled device in which the voltage on an insulated gate creates a conducting channel between source and drain. Because the gate sits on an insulator, it draws no DC current.

2 Why is gate current zero?

The gate is separated from the channel by 4.1 nm of silicon dioxide, an insulator. The gate is one plate of a capacitor, so it conducts only while its voltage is *changing*.

3 What is threshold voltage?

The gate–source voltage at which an inversion layer forms and the channel begins to conduct. About 0.50 V for NMOS and 0.51 V for PMOS in this process.

4 Difference between V_{GS} and V_{DS} in what they control?

V_{GS} sets *how much* current the device passes. V_{DS} sets *which region* it operates in — triode (resistor-like) or saturation (current-source-like).

5 How do you tell the source from the drain?

The circuit decides. For an NMOS the source is the lower-voltage terminal; for a PMOS it is the higher-voltage one. They are physically identical.

6 NMOS or PMOS — which is faster and why?

NMOS. Electron mobility (417 $cm^2/V{\cdot}s$) is about five times hole mobility (85 $cm^2/V{\cdot}s$), so a PMOS needs roughly five times the width for the same current.

7 What is transconductance?

$g_m = \partial I_D / \partial V_{GS}$, in siemens — how much drain current changes per volt of gate drive. It is the amplification mechanism of a MOSFET.

8 What is $R_{DS(on)}$?

The resistance of a fully-on MOSFET in triode. Here $R_{DS(on)}$ of M_P is 1.82 Ω, which is what sets the 182 mV dropout at 100 mA.

9 What is the body diode and why does it exist?

The PN junction between each source/drain region and the body. It exists unavoidably because you built n-type regions inside p-type silicon. It must be kept reverse-biased, which is why PMOS bodies tie to VIN and NMOS bodies to GND.

10 What makes a MOSFET behave like a diode?

Shorting gate to drain. That forces $V_{GS} = V_{DS}$, which guarantees saturation whenever it conducts, so it becomes a two-terminal element that conducts above V_{TH} and blocks below. This is separate from the body diode.

11 What is a current mirror?

A diode-connected device converts a reference current into a gate voltage; a second device sharing that gate voltage reproduces the current, scaled by the W/L ratio.

12 What is a differential pair?

Two matched transistors sharing a tail current source. The tail total is fixed; the input voltage difference decides how it splits. That is how analog subtraction is done.

13 What is negative feedback and why use it?

Feeding the output back so that it opposes its own change. It trades raw gain for accuracy, linearity, and low output impedance — all of which the LDO needs.

14 What does "DC" mean and why must this circuit be DC?

Constant, one-polarity voltage. MOSFETs are one-polarity devices with polarity-dependent body diodes; on a reversed half-cycle the wells forward-bias and the chip is destroyed.

15 What is ground?

The node we declare to be 0 V, because voltage only exists as a difference. All other node voltages in the circuit are stated relative to it.

16 What is dropout voltage?

The minimum $V_{IN} - V_{OUT}$ at which the regulator still regulates. Below it the pass device is fully on and the output simply follows the input minus $I \cdot R_{DS(on)}$.

17 What is quiescent current?

The current the regulator consumes for itself, independent of the load. About 9 µA at light load here, rising to ~29 µA at full load because of the adaptive bias.

18 What are line and load regulation?

Line regulation is ΔV_{OUT} per volt of input change; load regulation is ΔV_{OUT} per amp of load change. Both improve in direct proportion to loop gain. Here load regulation is ~4.2 mΩ.

19 What is phase margin and why 60°?

How much phase the loop has left before reaching −180° at unity gain. Below 45° the response rings badly; above about 75° it is over-damped and slow. 60° is the usual compromise.

20 What is PSRR?

Power Supply Rejection Ratio — how much input noise appears at the output. At low frequency it is set by loop gain; it degrades at high frequency as loop gain rolls off and the pass device's parasitic capacitance couples VIN straight through.

B. Fifteen questions on your specific circuit

1 Walk me through your circuit.

"V_{OUT} is compared with V_{REF} by a cross-coupled current-mirror OTA whose output V_G drives a PMOS pass device. C_M compensates the loop. A replica of the pass device senses the load current and raises the amplifier's tail current in proportion. Two capacitively-coupled class-AB boosters bypass the amplifier on fast transients to slew V_G." Then stop and let them pick a block.

2 Why is M_P a PMOS?

Because its source sits on VIN, the gate is driven *downwards* to turn it on, so no voltage above VIN is needed. An NMOS pass device would need $V_G \approx 2.25$ V, i.e. a charge pump, or it would suffer a 650 mV dropout.

3 How did you size M_P?

From the triode equation at the worst case — 100 mA at 200 mV V_{SD} with $V_{SG} = 1.6$ V — giving $W/L = 7093$, so W = 2.5 mm at L = 0.35 µm.

4 What do M_{P3} and M_{P4} do?

They are cross-coupled, providing a negative resistance that partially cancels the diode-connected loads M_{P1}/M_{P2}, boosting the first-stage gain without extra current or an extra pole. The gain boost is $1/(1 - g_{m3}/g_{m1})$.

5 What is the risk with M_{P3}/M_{P4}?

If $g_{m,MP3}$ reaches $g_{m,MP1}$ the positive feedback exceeds unity and nodes X and Y latch. I keep the ratio at 0.5 and verify across corners and temperature.

6 Why four input transistors instead of two?

M_{N3} and M_{N4} reach the output branches directly, in one device delay, while M_{N1}/M_{N2} go via a current mirror. It gives a fast path and a push–pull output stage. It does not, by itself, double the DC transconductance — the shared tail means each device carries half the current.

7 What does M_{N7}/M_{N8} do?

Differential-to-single-ended conversion. It folds the left branch's current into V_G so both halves of the differential signal contribute, and M_{N8} provides the pull-down on V_G.

8 Explain the adaptive bias block.

M_{P9} shares M_P's gate and source, so it carries $I_{LOAD}/5000$. Diode-connected M_{N10} converts that to V_{AB}, which drives M_{N9} in parallel with the fixed tail source. The amplifier's bias current therefore tracks the load: 2 µA light, 22 µA at 100 mA.

9 Why does adaptive bias improve stability rather than harm it?

The output pole is $g_{mp}/2\pi C_L$ and $g_{mp} \propto \sqrt{I_{LOAD}}$, so it moves to higher frequency as the load rises. Adaptive bias raises the loop's GBW at the same time, keeping the GBW-to-pole ratio, and therefore the phase margin, roughly constant.

10 Why does the SRE need two stages?

Sign first: each common-source stage inverts, and V_{OUT} falling must make V_G fall. One stage would give the wrong polarity and create runaway positive feedback. Two also provide the gain to turn a 50 mV dip into a volt-scale gate swing.

11 What are M_{P12} to M_{P15}?

Pseudo-resistors — diode-connected PMOS biased at essentially zero current, so their incremental resistance is in the gigaohm range. They set each booster gate's DC level without shunting the AC signal, giving a high-pass corner around 800 Hz with 200 fF coupling capacitors.

12 Why is C_M connected from V_G to V_{OUT} and not to ground?

Across the gain stage it is Miller-multiplied by $(1+A_2)$, so 5 pF does the job of 200 pF — a 40× area saving — and it splits the poles. It also feeds the output dip forward to the gate with literally zero delay.

13 What are M_{P7} and M_{P8}?

A diode-connected minimum-load bleeder drawing about 3 µA. They keep the loop at a defined operating point at zero load, discharge the output, and give a defined start-up state. There is no tap between them, so they are not a feedback divider.

14 What sets your output voltage?

V_{REF} directly. V_{OUT} is fed back to the amplifier with no divider, so this is unity-gain feedback and $V_{OUT} = V_{REF} = 1.6$ V. Accurate and low-power, but fixed — an adjustable version needs a divider.

15 What is your loop gain and what does it buy you?

89 dB at light load, 70 dB at full load. At full load that gives a closed-loop output resistance of 4.2 mΩ, so the output moves about 0.4 mV over the full load range.

C. Ten questions designed to catch you out

1 Your paper claims 20 mV undershoot for a 100 mA step on 100 pF. Prove it.

"I can't, and I don't claim it. $\Delta V \approx \Delta I \cdot t_{resp}/C_L$ requires $t_{resp} \approx 20$ ps for that number, which is not achievable. With my 6 ns response time, 100 pF supports a fast step of about 0.8 mA within 50 mV; a genuine 100 mA step needs roughly 6 nF, or a load edge slow enough for the loop to track. The 100 mA figure in my design is a DC rating." **This is the highest-value answer in the whole document — an examiner who asks this is testing whether you reproduce claims or verify them.**

2 You said the dual input pairs double the transconductance. Do they?

No. The four devices share one tail, so each carries $I_{tail}/4$ and the total G_m comes out similar to a plain pair at the same tail current. What they actually buy is a fast direct path to the output that skips the mirror pole, and a push–pull output. Any G_m gain comes from the mirror ratio B and the cross-coupling factor k, not from the doubling itself.

3 Which load condition is worst for stability, and why is that surprising?

Light load. Intuition says heavy load is harder, but $g_{mp} \propto \sqrt{I_{LOAD}}$, so at light load the output pole falls to within a factor of two of the unity-gain frequency. At heavy load it is 20× away.

4 Adaptive bias senses V_G. Isn't that circular?

It is a legitimate observation. V_G is the amplifier's own output, so the extra bias only arrives *after* V_G has moved — which is after the amplifier needed it. Adaptive bias is therefore a steady-state optimisation, not a transient one, and that is exactly why the SRE block, which is AC-coupled directly from V_{OUT}, is also required.

5 Node A in your SRE has no DC path. What sets its voltage?

It self-biases at the level where M_{P10}'s and M_{N11}'s subthreshold currents are equal. That is a standard low-power technique but it is corner-sensitive, since subthreshold current depends exponentially on V_{TH}. I verify node A's DC level across all corners and temperatures, and if it drifts too far I would define it with a weak pseudo-resistor to a mid-rail bias.

6 Where is the right-half-plane zero, and what did you do about it?

At $g_{mp}/2\pi C_M$ — 94 MHz at light load, well above GBW, but it moves down as the load falls further. The schematic has no nulling resistor, so it is uncompensated. Adding $R_Z \geq 1/g_{mp} \approx 337\ \Omega$ in series with C_M moves it into the left half plane, where it improves phase margin instead.

7 What happens if V_{OUT} is pulled above V_{IN}?

M_P's drain–body diode forward-biases and current flows backwards from output to input, regardless of V_G. The gate has no control over it because it is a parasitic junction, not the channel. Preventing it needs back-to-back devices or a reverse-current detector, neither of which is in this design.

8 What is the quiescent current at exactly zero load?

About 9 µA, of which 3 µA is the M_{P7}/M_{P8} bleeder — so a third of the standby current is deliberately wasted to keep the loop well-behaved at zero load. That is a real trade-off, and one worth revisiting if standby is the priority.

9 Why is there no current limit, and what would you add?

It is a genuine omission. On an output short the loop drives V_G fully down and the pass device dissipates over a watt. I would add a second replica of M_P feeding a comparator against a reference, whose output pulls V_G up to fold back the current — see my improvement list.

10 Your hand calculations use long-channel equations for a 0.35 µm device. Is that valid?

Approximately, and I know its limits. Velocity saturation and mobility degradation mean the square-law under-predicts the width needed, typically by 20–50 %. I use the hand equations to choose the topology and the starting point, and the simulator with BSIM models to size the devices.

D. Five mistakes students make explaining this circuit

1. **Calling M_{P7}/M_{P8} the feedback divider.** Look for the tap. There isn't one, so $V_{OUT} = V_{REF}$ and they are a bleeder. Getting this right immediately signals that you read the schematic.
2. **Saying the SRE "adds gain".** It provides *current* to slew a capacitive node. Its whole purpose is charge delivery, not amplification. The gain in the boosters is a means, not the end.
3. **Confusing the body diode with the diode connection.** One is parasitic and gate-independent; the other is deliberate and gate-defined. Mixing them up is the fastest way to lose a diode question.
4. **Claiming heavy load is the stability worst case.** It is the opposite, and the $\sqrt{I_{LOAD}}$ relationship is why.
5. **Quoting a transient number you cannot derive.** Any published figure you repeat, an examiner may ask you to justify. Derive it, or state it as a reference result and say what your own analysis gives.

PART 17

Making this design your own

Your professor asked you to turn a basic LDO into a workable, clean circuit. Understanding it is step one; this is step two. Each item below is a real, bounded contribution you could implement, simulate and defend — ordered by value-per-effort.

#	IMPROVEMENT	WHAT IT FIXES	EFFORT
1	**Nulling resistor in series with C_M** — a PMOS in triode sized for $R \approx 1/g_{mp}$, ideally one whose gate is driven by V_{AB} so R tracks the load	Moves the RHP zero to the left half plane, adding phase margin. The load-tracking version is a genuinely publishable refinement	1–2 devices. Very high value
2	**Current limit with foldback** — a second replica of M_P, a comparator against a reference, output pulling V_G up	Short-circuit survival. Without it the part is not usable in any real product	5–8 devices. High value
3	**Start-up circuit** — inject current into the bias node until the loop is alive, then disable itself	Eliminates the all-off stable state. Demonstrate with a slow V_{IN} ramp from 0 V	3–4 devices. High value, easy to demo
4	**Define node A's DC level** — a weak pseudo-resistor from A to a mid-rail bias, or a diode-connected replica pair	Removes the corner sensitivity you identified. Show before/after corner sweeps of node A	1–2 devices. Good value
5	**Cascode the pass device's gate drive, or add a PSRR-enhancing feed-forward path**	High-frequency PSRR, which is the weakest headline spec of this topology	Moderate. Strong differentiator if you measure it
6	**Optional resistive or diode divider with a switchable tap**	Makes V_{OUT} selectable (1.2/1.6/1.8 V) instead of fixed at V_{REF}	Low. Useful if your brief asks for adjustability
7	**Latch-up detector on the cross-coupled pair** — sense X and Y both at rails and reset the node	Turns your best-known risk into a handled fault. Unusual and memorable	Moderate. Excellent viva material
8	**Right-size the load-step spec** — publish the ΔI–C_L–ΔV trade curve from Part 13.7 instead of a single claimed number	Turns the topology's honest limit into your own analytical contribution	Analysis only. Highest value per hour spent

IF YOU WANT ONE THING THAT LIFTS THIS FROM "A WORKING CIRCUIT" TO ORIGINAL WORK

Item 8 combined with item 1. The trade-curve analysis is something almost no student paper does: a plot of achievable undershoot against C_L and loop response time, with your measured simulation points on it, showing exactly where the capacitorless approach stops being viable. It costs no extra silicon, it is rigorously derivable, and it demonstrates the thing examiners and reviewers actually value — that you know what your circuit *cannot* do and why.

Item 1's load-tracking nulling resistor is the natural companion: a physical modification, easy to simulate, and it uses V_{AB}, a signal your circuit already generates. That reuse is the kind of detail that reads as design insight rather than bolt-on.

The simulation set to run, in order

1. **DC operating point** — every device's region, I_D, V_{GS}, V_{DS}. Do this first, always.
2. **DC sweep of V_{IN}** 0 → 2 V, plotting V_{OUT} — gives line regulation and the dropout point.

3. **DC sweep of I_{LOAD}** 1 µA → 100 mA — gives load regulation; confirm the 0.4 mV prediction.
4. **STB (stability) analysis** at I_{LOAD} = 1 µA, 100 µA, 1 mA, 10 mA, 100 mA — plot loop gain and phase margin against load. Confirm light load is the worst case.
5. **Transient load step** at several edge rates, plotting V_{OUT}, V_G, V_{AB}, node A and the booster currents on one axis — this is the plot that shows all three response mechanisms doing their jobs on their own timescales, and it is the single best figure for your report.
6. **AC PSRR** — inject on V_{IN}, measure at V_{OUT}, 1 Hz to 100 MHz.
7. **Corner and temperature sweep** — five process corners × −40/27/125 °C, checking specifically for cross-coupled latch-up and node A's DC level.
8. **Start-up transient** from V_{IN} = 0 with realistic ramp times.

Sources for the process data. Threshold voltages, mobilities, oxide thickness and transconductance parameters are taken from a published TSMC 0.18 µm WAT parameter summary; topology context for adaptively-biased capacitorless LDOs with capacitively-coupled slew-rate boosters comes from the open-access literature listed in the conversation alongside this page.

Every number in Part 13 was computed from the process constants in Part 13.0 and can be reproduced on paper. Where a result contradicts a commonly quoted figure — notably the transient droop in Part 13.7 — the arithmetic is shown so you can check it yourself rather than take it on trust. Device sizes are hand-calculation starting points using long-channel equations; final sizing must come from simulation with the foundry's BSIM models.